

5

Wave Phenomena and Stationary Waves

In this chapter, we will learn about

- representing waves using wavefronts
- the four phenomena of waves—reflection, refraction, diffraction and interference
- the formation and characteristics of stationary waves

5.1 Water waves

Video

→ This video demonstrates the Hands-on activity.

Flipped classroom

Watch a video and answer the questions.

Hands-on activity

→ Worksheet is available on OUP web.

Light and shadow

What you need



What to do



Fill the transparent packaging with water and hold it over a plain, flat surface. Shine the lamp through the packaging from above to project a pattern on the surface. Observe the bright lines and dark lines formed in the pattern.

What you find

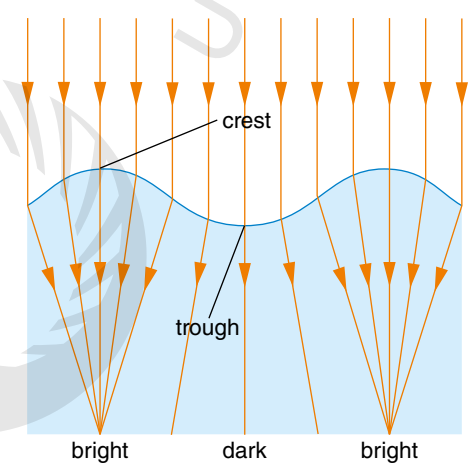
Which parts of the packaging correspond to the bright lines and which parts to the dark lines? See Fig 5.1a.

In the above activity, the packaging resembles the shape of a water wave. A pattern of bright and dark lines is formed when light passes through the packaging filled with water (Fig 5.1a). A similar pattern can be seen when light passes through a water wave. This is because the crests and troughs act like a series of convex and concave lenses (Fig 5.1b).

A water wave is a transverse wave. Water molecules vibrate perpendicularly to the direction of travel of the wave. ▶



Fig 5.1a *Light passes through the packaging filled with water.*



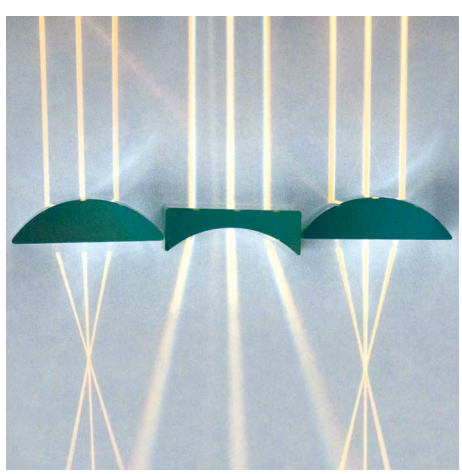


Fig 5.1b *Bright and dark lines are formed when light passes through a water wave.*

This allows us to observe water waves clearly. In this chapter, we will use water waves to study some common wave phenomena.

Know more

Ripple tank
A ripple tank has some special features.
▸ p.99

1 Observing water waves in a ripple tank

A **ripple tank** (Fig 5.1c) is a useful experimental set-up for studying water waves. It consists of a shallow transparent tray filled with water and a vibrating source. When the water surface is repeatedly hit by the vibrating part, water waves are produced. A lamp can be used to project the wave pattern onto a flat surface.

In the experiment, Ts may record the wave motion with a smartphone using slow motion mode. To allow Ss better judge the speed of the wave, Ts may (i) mark position of 5 cm, 10 cm, 15 cm, etc. from the bar on the transparent tray and (ii) include a stopwatch when recording the video.

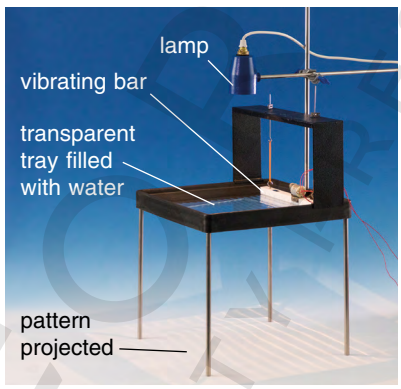


Fig 5.1c *A ripple tank.*

Experiment 5a Producing waves in a ripple tank

Video

→ This video shows how straight and circular waves are produced in the ripple tank.

1 Set up a ripple tank (Fig a).

2 Tap the water surface with a bar (Fig b). Observe the water wave produced.

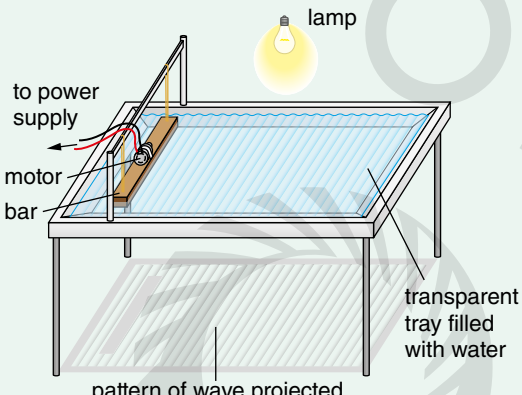


Fig a

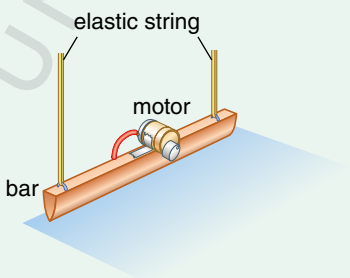


Fig b

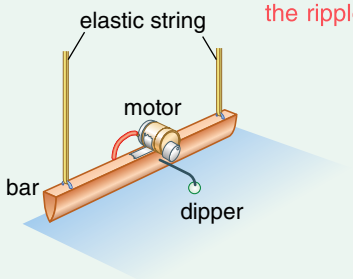


Fig c

3 Increase the vibrating frequency of the bar and see how the wavelength and the speed of the water wave change.

4 Repeat the experiment by using a dipper to tap the water surface (Fig c).

Ts may ask Ss to observe the shape of the wave produced by a dipper. It gives waves of circular shape, which implies that the water wave speed is constant in all directions.

Discussion

1 Do the water waves travel at a constant speed?

2 What happens to the wavelength if you increase the vibrating frequency?

ripple tank 水波槽

In Experiment 5a, a **straight wave**, also called a **plane wave**, is produced by a vibrating bar (Fig 5.1d). A **circular wave** is produced by a vibrating dipper (Fig 5.1e).

In Figure 5.1e, the elastic string is shortened so that only the dipper taps the water surface.

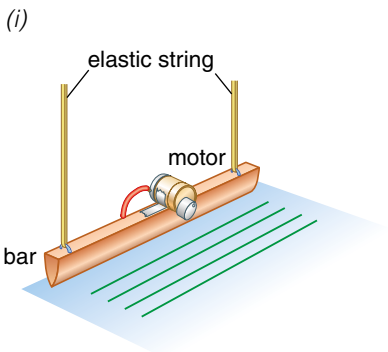


Fig 5.1d *(i) A straight wave and (ii) its projected pattern.*

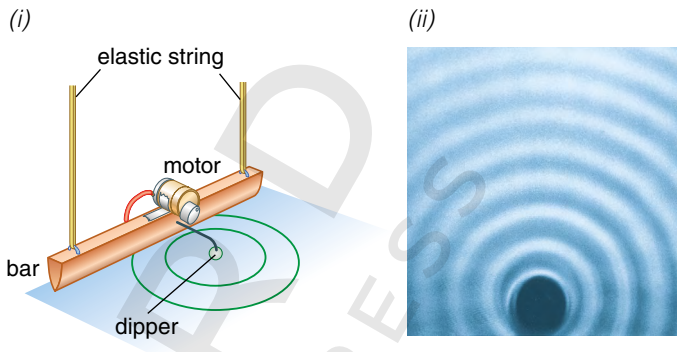


Fig 5.1e *(i) A circular wave and (ii) its projected pattern.*

A dark line is a line of troughs.

In the projected wave patterns, a bright line is a line of crests. These lines can be taken as the **wavefronts** of the wave. A wavefront is the line joining the neighbouring particles vibrating **in phase**. It is always **perpendicular** to the direction the wave travels.

In some cases, lines of troughs are also drawn.

In drawing a wave pattern, we usually only draw the wavefronts of crests. In this case, the distance between two successive wavefronts equals **one wavelength** (Fig 5.1f).

Sometimes, the direction of travel is also called a ray.

In some Physics curricula (e.g. GCSE) of other countries, the term 'ray' is used to refer to the travelling direction of the wave.

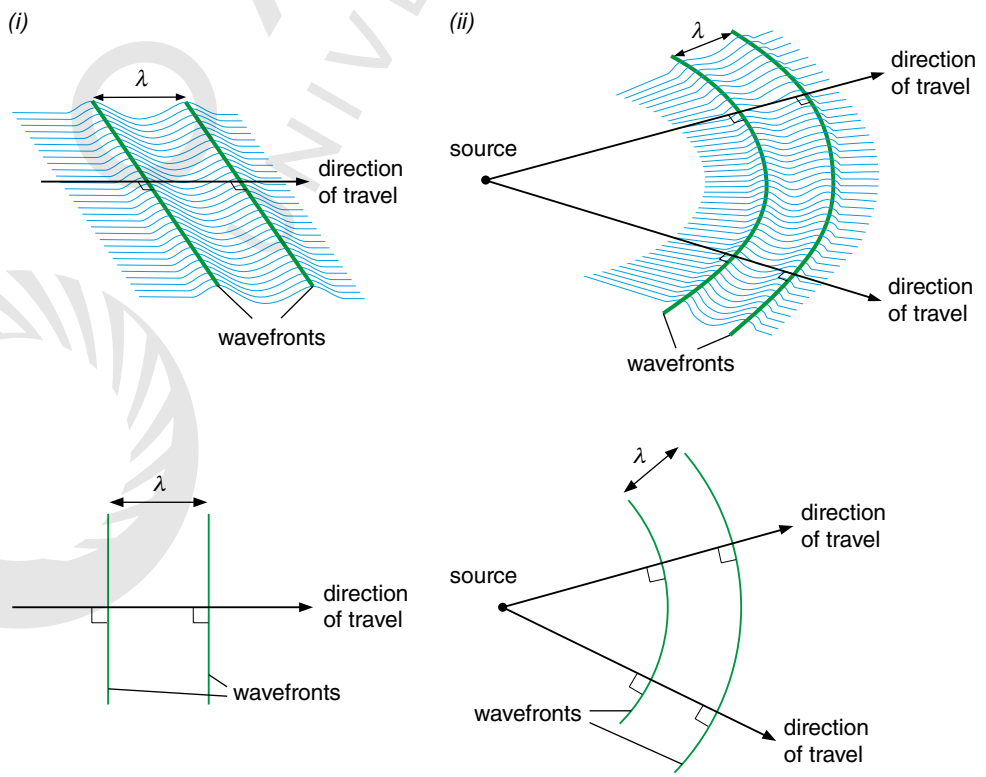


Fig 5.1f *Wavefronts and direction of travel.*

straight wave 直線波　plane wave 平面波　circular wave 圓形波　wavefront 波陣面

2 Wavelength, frequency and speed

We can measure the wavelength of a water wave by putting a transparent ruler in a ripple tank. The distance between two successive bright lines (or two successive dark lines) equals one wavelength, which can be read directly from the projected image of the ruler (Fig 5.1g).

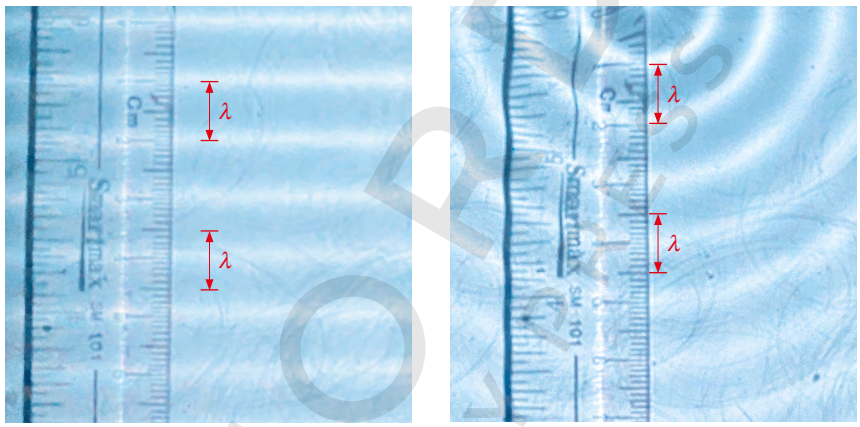


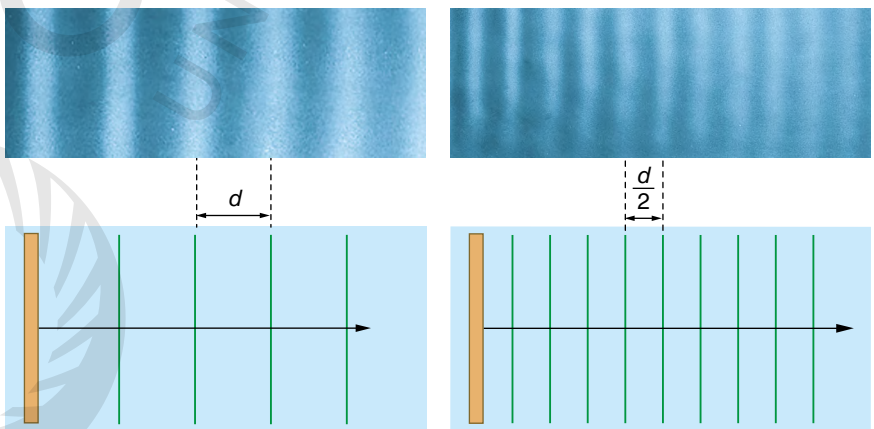
Fig 5.1g *Measuring the wavelengths of water waves produced in a ripple tank.*

A circular wave implies that the wave moves away from the source with the same speed **in all directions**.

Recall that a transverse wave travels at a constant speed along a uniform spring (p.10). A similar result can be observed in Experiment 5a. When the water depth is uniform, a water wave travels at a constant speed v.

If f is constant, evenly spaced wavefronts implies that the wave travels at a constant speed.

The frequency f of a water wave is equal to that of the vibrating source. If f and the water depth are constant, by $v = f\lambda$, λ will remain constant, so the lines in the wave pattern will be evenly spaced. If f increases, λ will decrease (Fig 5.1h).



(i) A water wave with frequency f. *(ii) A water wave with frequency 2f.*

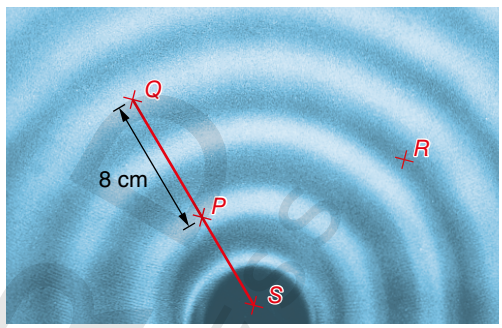
Fig 5.1h *Wavelength decreases as frequency increases.*

Ss can verify this at home. Dip the water surface using a finger. Note the wavelength of the water waves produced. Then increase the dipping frequency to see how the wavelength changes.

With a constant water depth, a water wave travels at a constant speed. In this case, $f\uparrow \Rightarrow \lambda\downarrow$

Example 1 | Wave pattern produced in a ripple tank

The figure shows a wave pattern formed by a ripple tank with a constant water depth. Point S is the position of the dipper.



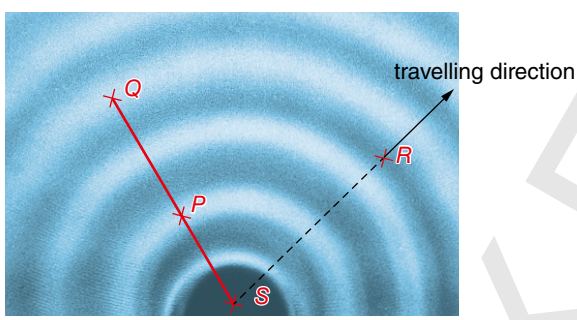
(a) Draw an arrow to show the travelling direction of the wave at R.

(b) The frequency of the dipper is 6 Hz. Find the speed of the water wave.

(c) The vibrating frequency of the dipper is now doubled. How will the wave speed and the wavelength change? Explain briefly.

Solution

(a)



(b) Wavelength $\lambda = \frac{8}{2} = 4$ cm $= 0.04$ m

$v = f\lambda = 6 \times 0.04 = 0.24$ m s^{-1}

(c) Since the water depth is unchanged, the wave speed remains unchanged.

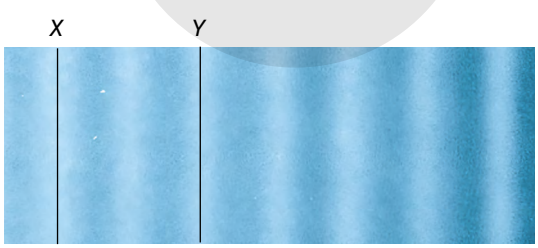
By $v = f\lambda$, for constant v, λ will be reduced by half as f is doubled.

Quick check 1

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 A bar vibrates at a constant frequency f in a ripple tank to produce a straight wave as shown. The wave has a constant wavelength λ. Determine whether the following statements are true.



(a) The distance between lines X and Y is 4λ. (T/F)

(b) The wave travels at a constant speed. (T/F)

(c) The travelling direction of the wave is perpendicular to the wavefronts. (T/F)

(d) The water depth is constant. (T/F)

(e) If f increases, the wave speed increases. (T/F)

(f) If f increases, the wavelength decreases. (T/F)

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Exercise 5.1

★**1** A vibrating source produces a water wave in a ripple tank. If the vibrating frequency of the source is doubled, which of the following statements is/are correct?

(1) The wave speed is doubled.

(2) The wave frequency is doubled.

(3) The wavelength is unchanged.

A (1) only B (2) only

C (1) and (2) only D (1), (2) and (3)

★**2** The wave pattern projected from a ripple tank at an instant is shown below (Fig a). The bright lines are evenly spaced.

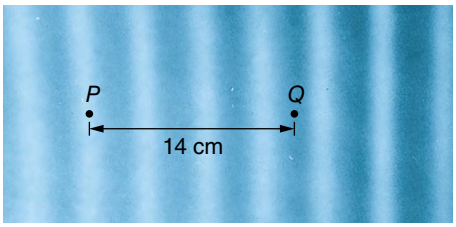


Fig a

Which of the following statements is/are correct?

(1) Point P is at a crest at this instant.

(2) The wavelength of the wave is 3.5 cm.

(3) The wave travels at a constant speed.

A (1) only

B (1) and (3) only

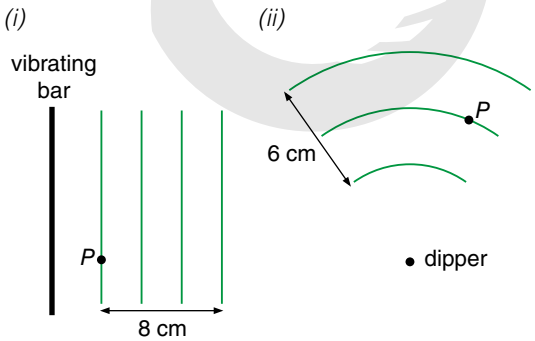
C (2) and (3) only

D (1), (2) and (3)

3 Water waves are produced in a ripple tank. The green lines represent wave crests. Each wave has a constant wavelength. In each case,

(a) find the wavelength of the wave;

(b) draw an arrow to show the travelling direction of the wave at position P.



★**4** A water wave of wavelength 1 cm is produced in a ripple tank at 10 Hz. The depth of the water in the tank remains unchanged. Figure b shows the wave pattern observed at time $t = 0$. The green lines represent wave crests.

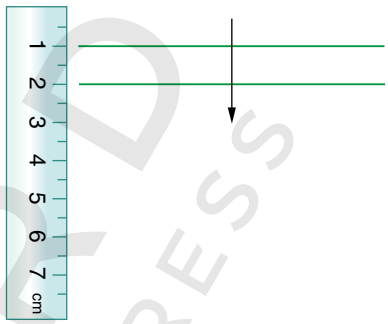


Fig b

(a) What is the wave speed?

(b) What is the distance travelled by the wave in 0.4 s?

(c) Draw the wavefronts observed at $t = 0.4$ s in Figure b.

(d) If the frequency of the wave is reduced to 4 Hz, how do the wavelength and wave speed change?

★**5** A straight water wave is travelling to the left in a ripple tank. Figure c shows the wave pattern at a certain instant. The green lines represent wave crests.

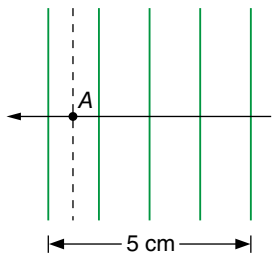


Fig c

(a) What is the wavelength of the wave?

(b) A ball is floating at point A, which is in the middle of two green lines at the instant shown.

(i) Is the ball at the crest, the trough or the equilibrium position of the water wave?

(ii) Describe the motion of the ball at this instant.

(c) The wave speed is 4 cm s^{-1}. What is the frequency of the wave?

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5.2 Reflection and refraction

In Book 3A we learned that a light ray is reflected when it hits a mirror and is refracted when it travels from one medium to another. In fact, all waves can be reflected or refracted. In this part, we will use water waves to study the reflection and refraction of waves.

DSE goal

Realize the reflection of waves at plane barrier.

Draw wavefront diagrams to show reflection.

DSE exam

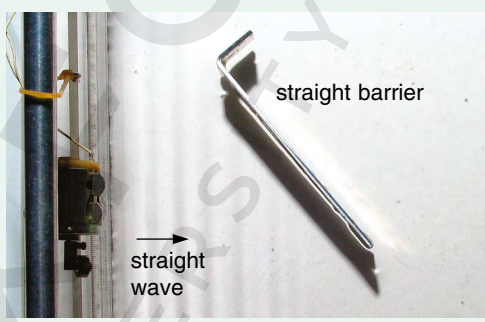
14(1A)Q13

1 Reflection

Experiment 5b / Reflection of water waves

1 Set up a ripple tank.

2 Send a straight wave pulse towards a straight barrier.



Record a video of the wave motion with a smartphone by using slow motion mode. Ask Ss to observe the wave motion again with the video.

3 Repeat the experiment with a series of straight waves.

4 Repeat steps 2 and 3 using a dipper instead of a bar to produce circular waves.

Discussion

Does the wavelength change after the wave hits the barrier?

Simulation Video

→ This simulation shows reflection of straight and circular water waves by straight and curved reflectors. It is available on OUP web in PC version (.exe) only.

→ This video shows the reflection of water waves by straight and curved barriers in the ripple tank.

Simulation

→ This simulation shows the reflection of straight and circular water waves by a straight reflector. Direction and frequency of the incident wave can be varied.

a Straight waves

In Experiment 5b, when a straight wave pulse or a series of straight waves hit a barrier, they bounce off the barrier (Figs 5.2a and 5.2b).

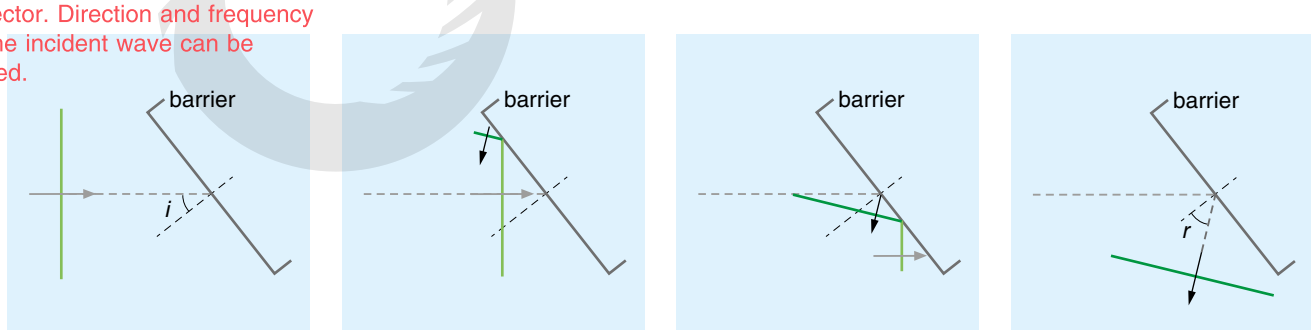


Fig 5.2a *The reflection of a straight wave pulse.*

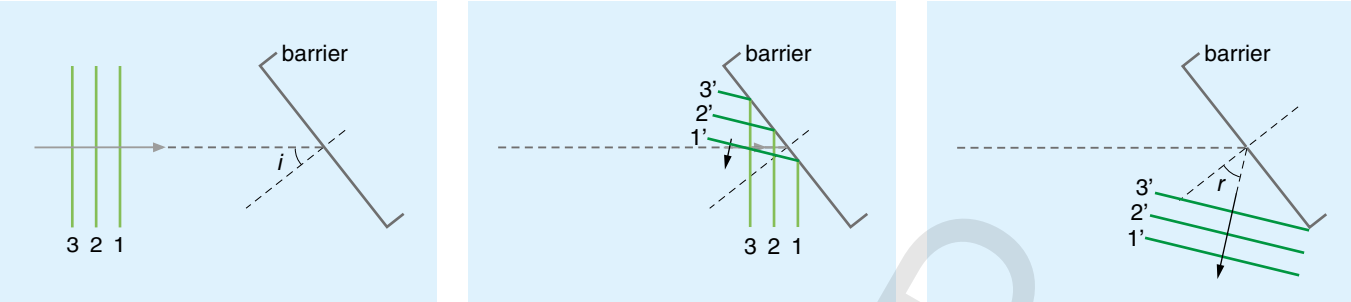


Fig 5.2b *The reflection of a series of straight waves.*

Cross-link

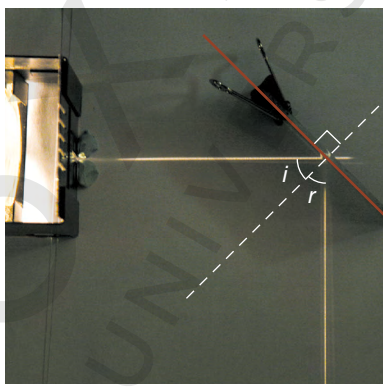
You learned the laws of reflection in Book 3A Ch 1.

▶ Similar to the reflection of light rays, the reflection of all waves obey the laws of reflection (Fig 5.2c):

angle of reflection ····▸ $r = i$ ◂···· angle of incidence

Note that i is the angle between the **travelling direction** of the incident wave and the normal; r is the angle between the travelling direction of the reflected wave and the normal.

(i)



(ii)

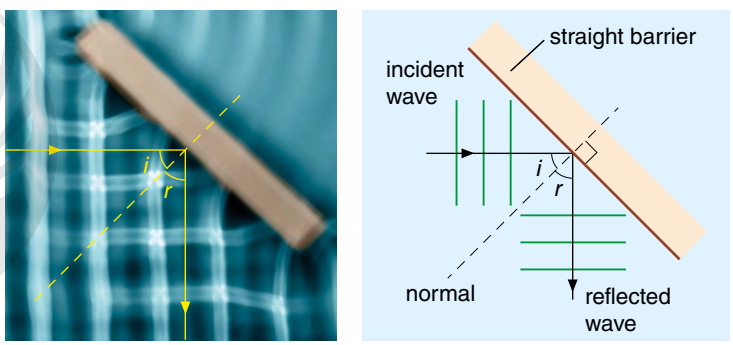


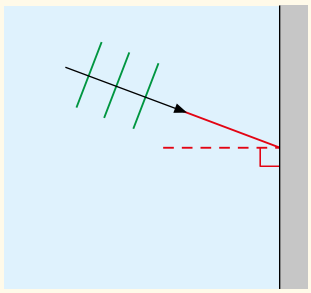
Fig 5.2c *The reflection of (i) a light ray and (ii) a water wave. In both cases, r = i.*

The frequency f, wave speed v and wavelength λ of the wave remain unchanged after reflection.

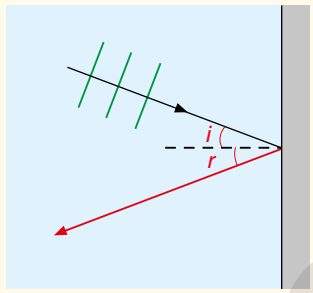
→ Teaching notes and exercises are available on OUP web.

Skill
Drawing

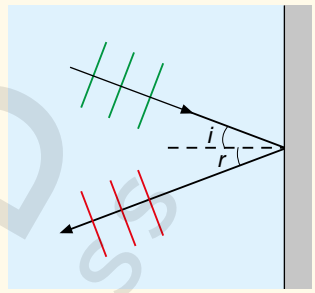
Drawing reflected wavefronts of straight waves

① 

Extend the line indicating the direction of travel of the incident wave. Draw the normal at the point where the line meets the barrier.

② 

Draw an arrow to indicate the direction of travel of the reflected wave with $r = i$.

③ 

Draw the reflected wavefronts perpendicular to the new arrow in ②. The separation between successive wavefronts should be equal to that between the incident wavefronts.

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 A straight wave is reflected by a straight barrier. Determine whether the following statements are true.

(a) The angle of incidence is equal to the angle between the wavefront and the normal. (T/F)

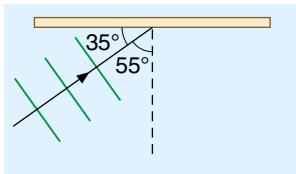
(b) Due to energy loss during reflection, the wave speed decreases after reflection. (TF)

(c) The wavelength of the wave decreases after reflection. (T/F)

(d) The frequency of the wave remains unchanged after reflection. (T/F)

2 A water wave hits a straight barrier as shown. What is the angle of reflection?

A 35°
B 55°
C 70°
D 125°

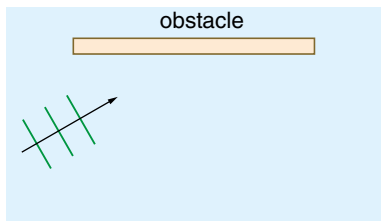


Learn by practice 1

1 A straight wave approaches a straight obstacle as shown.

(a) Sketch the reflected wave.

(b) Label the angle of incidence i and angle of reflection r.



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b Circular waves

Figure 5.2d shows the reflection of a circular wave by a straight barrier. The reflected wave appears to originate from point *I* behind the barrier (Fig 5.2e). *I* is as far away from the barrier as the point source *O* is.

Note that the angle of reflection *r* is always equal to the angle of incidence *i* at all points of incidence (Fig 5.2f).

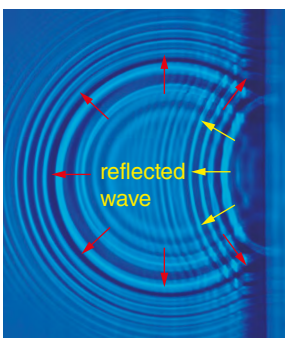


Fig 5.2d *Circular wave reflected by a straight barrier.*

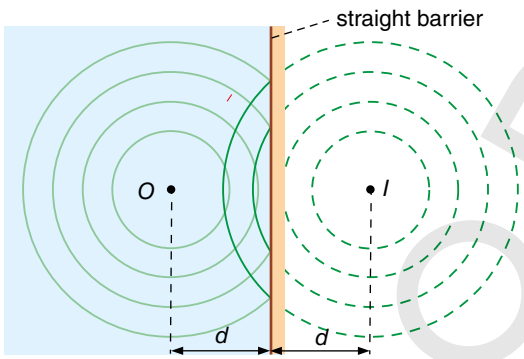


Fig 5.2e *The reflected wave appears to come from a point I behind the barrier.*

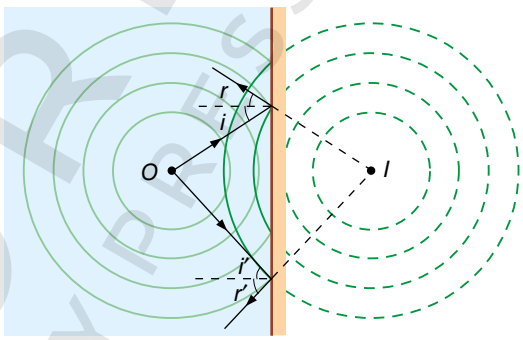


Fig 5.2f *Angle of reflection r equals angle of incidence i at all points of incidence.*

Google Forms Microsoft Forms

Quick check 3

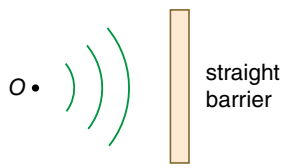
→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) When a circular wave hits a barrier, the angle of reflection may not be equal to the angle of incidence. (T/F)

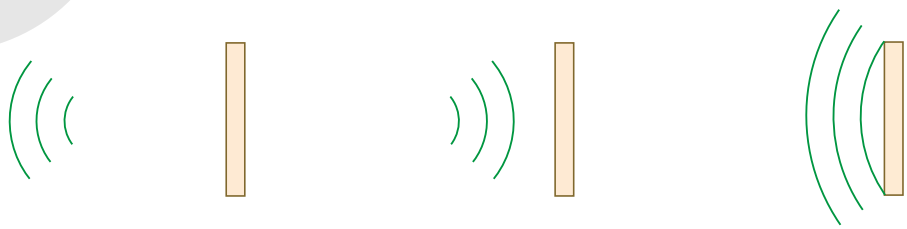
(b) When a circular wave is reflected by a straight barrier, its wavelength remains unchanged. (T/F)

2 Point source *O* produces a circular wave.



Which of the following diagrams best shows the wave reflected by the straight barrier?

A B C



DSE goal

Realize the refraction of waves across plane boundary.

Examine the change in wave speeds during refraction.

Draw wavefront diagrams to show refraction.

DSE exam

12(1A)Q17, 17(1A)Q16, 19(1B)Q5, 21(1B)Q6(a), 22(1A)Q17

Record a video of the wave motion with a smartphone in slow motion mode. Observe the wave motion again with the video and see whether there is any change in the wave when it enters the shallow region.

Simulation Video

→ This simulation shows the refraction of straight water waves when they travel from deep to shallow water at different angles. It is available on OUP web in PC version (.exe) only.

→ This video demonstrates Expt 5c.

2 Refraction

When light travels from one medium to another at an angle and has a change in speed, it undergoes refraction. A water wave behaves similarly.

Experiment 5c / Refraction of water waves

1 Set up a ripple tank. Compare the speeds of the water waves when the water depth is (a) 0.5 cm and (b) 3 cm.

2 Put a Perspex sheet in the tank to create a region of shallow water (the water should just cover the Perspex sheet).

3 Send a straight wave towards the Perspex sheet perpendicularly (Fig a). Observe any change in the wave when it enters the shallow region.

4 Adjust the Perspex sheet so that the wave passes over it at an angle (Fig b). Observe any change in the wave when it enters the shallow region.

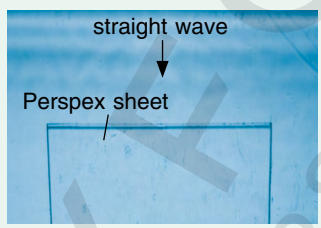


Fig a

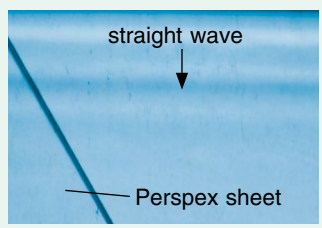


Fig b

Discussion

Do the wavelength and the travelling direction change when the wave enters the shallow region (a) perpendicularly, (b) at an angle?

a Change in wavelength and wave speed

In step 1 of Experiment 5c, when the water depth increases, the speed v of the water wave increases. In step 3, when the water wave enters the shallow region perpendicularly, its wavelength λ decreases (Fig 5.2g). This is because v decreases when the water wave enters the shallow region. Since the frequency f of the wave remains unchanged, by $v = f\lambda$, λ decreases when v decreases.

▶ The frequency remains unchanged because the wave in both deep and shallow regions come from the same source.

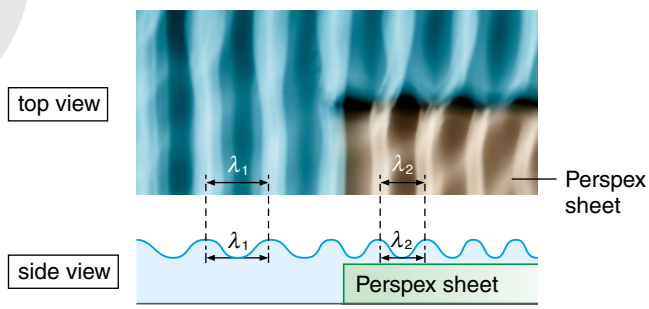
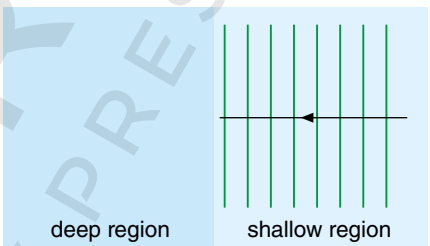


Fig 5.2g *Wavelength decreases as the wave enters the shallow region.*

When a water wave crosses from a deep to a shallow region,
- **its frequency f remains unchanged;**
- **its wavelength λ decreases;**
- **its wave speed v decreases.**

Example 2 | A wave entering a deep region

A straight water wave enters a deep region from a shallow region as shown.



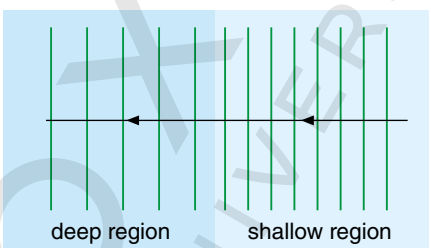
(a) Does the wave travel faster or slower in the deep region?

(b) Draw the wave pattern in the deep region.

Solution

(a) Faster

(b)

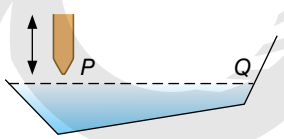


Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

(For Q1–2.) A vibrator generates a straight wave in a tilted ripple tank. Consider the wave travelling from P to Q.



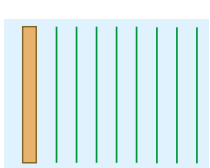
1 *True or false*:

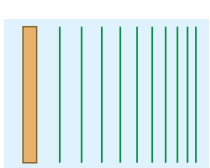
(a) The wave speed decreases. (T/F)

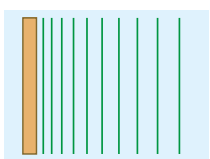
(b) The wave frequency decreases. (T/F)

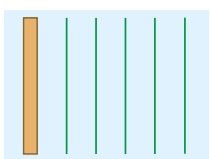
(c) The wavelength decreases. (T/F)

2 Which of the following wave patterns will you see?

A 

B 

C 

D 

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Simulation

→ This simulation shows the refraction of straight water waves across a straight boundary of two media. Direction and wavelength of the incident wave, and the refractive indices of the media can be varied.

b Change in wave direction

In Experiment 5c, when a water wave travels from a deep region to a shallow region at an angle, its travelling direction changes, i.e. the wave is refracted. This is similar to the refraction of light (Fig 5.2h).

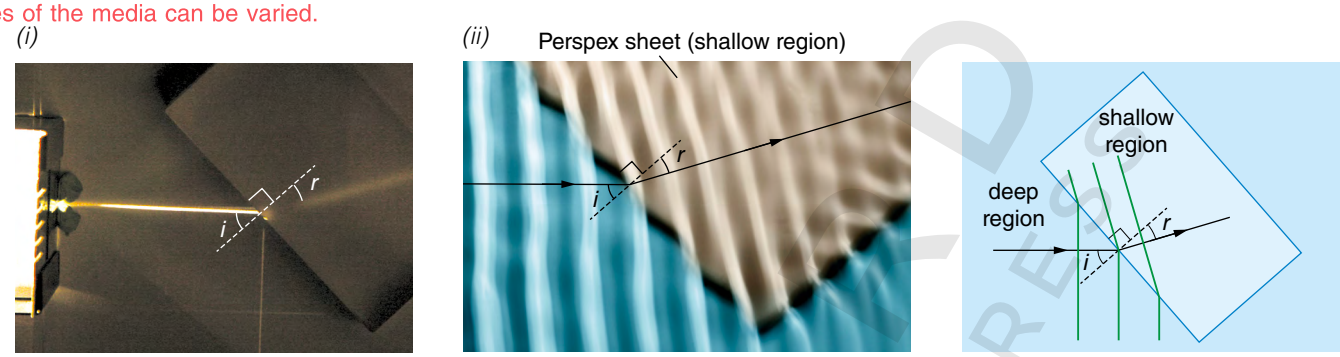


Fig 5.2h *Refraction of (i) a light ray and (ii) a water wave.*

A water wave bends in the same way as light.

1 When a wave travels from a deep region to a shallow region, it slows down after crossing a boundary and bends towards the normal, i.e. $r < i$.

2 When a wave travels from a shallow region to a deep region, it speeds up after crossing a boundary and bends away from the normal, i.e. $r > i$.

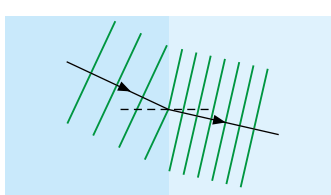
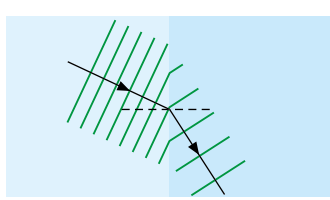
3 When a wave crosses a boundary perpendicularly, it does not bend.

▶ r is the angle between the travelling direction of the refracted wave and the normal.

▶ After crossing the boundary between two media at an angle, all waves bend towards the normal when they slow down. They bend away from the normal when they speed up.

Refraction is the change in travelling direction of a wave due to a change in wave speed.

Table 5.2a summarizes how water waves change during refraction.

	From a deep region to a shallow region	From a shallow region to a deep region
Frequency f	unchanged	
Wave speed v	decrease	increase
Wavelength λ	decrease	increase
Travelling direction	bend towards the normal 	bend away from the normal 

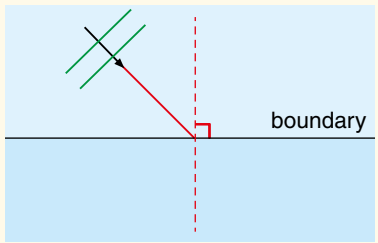
Strictly speaking, refraction involves a change in travelling direction. When a wave hits a boundary normally, there is no change in travelling direction and this should not be regarded as refraction.

Table 5.2a *Refraction of water wave.*

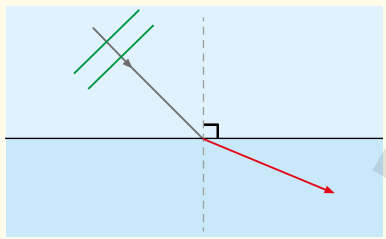
→ Teaching notes and exercises are available on OUP web.

Skill Drawing

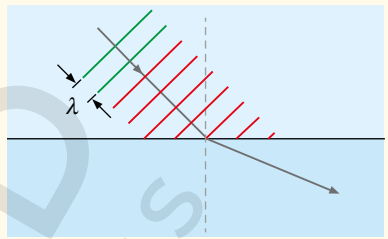
Drawing refracted wavefronts of straight waves

① 

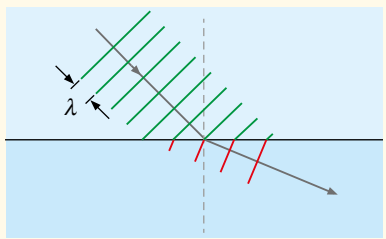
Extend the arrow showing the direction of the incident wave (the incident ray) to meet the boundary. Draw a normal at the meeting point.

② 

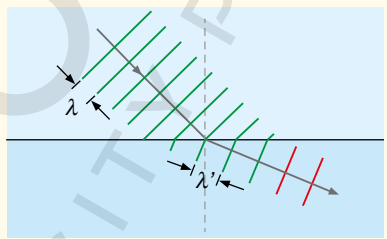
Draw the direction of the refracted wave (the refracted ray) according to the change in water depth across the boundary.

③ 

Draw the remaining incident wavefronts. λ (the separation between successive wavefronts) should remain the same.

④ 

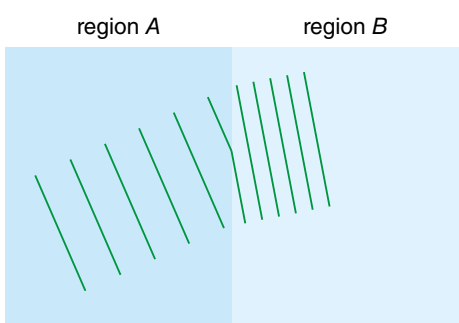
At the points where the incident wavefronts meet the boundary, draw the refracted wavefronts. The refracted wavefronts should be perpendicular to the refracted ray.

⑤ 

With λ' remaining constant, draw the remaining refracted wavefronts.

Example 3 | Refraction of a wave

A straight water wave travels from region A to region B as shown.



(a) When the wave crosses the boundary,
  (i) how does its frequency change?
  (ii) does it bend towards or away from the normal?

(b) Which region is deeper?

Solution

(a) (i) Its frequency does not change.

(ii) It bends towards the normal.

λ is larger in A. ▶

(b) Region A

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Know more

Waves approaching the shore

As waves approach the shore, the wavefronts become parallel to the shoreline (Fig a). Why?

Water is shallower near the shore (Fig b). As a wave approach the shore, it undergoes refraction and bends gradually until it travels along the normal to the shoreline (Fig c). Hence, the wavefronts become parallel to the shoreline.



Fig a

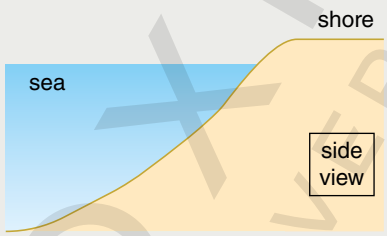


Fig b

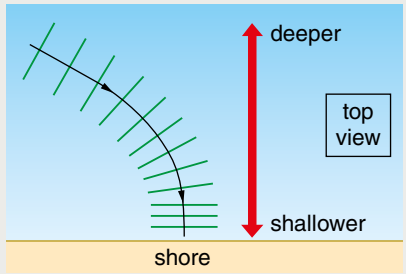


Fig c

Erosion of headlands

Many spectacular coastal rock features are often found at headlands protruding into the ocean (Fig d). This is partly because the erosion caused by the waves is stronger on the headlands.

When ocean waves from deep water approach the shore, the water becomes shallower and the waves refract. When these waves refract, their energy is concentrated around the headland (Fig e), causing particularly rapid erosion if weaknesses such as faults and fractures are present in the rocks.



Fig d

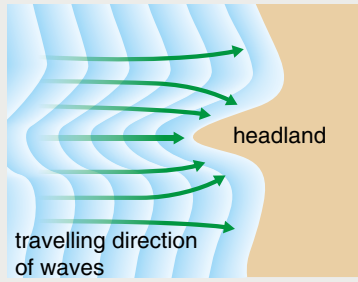


Fig e

Quick check 5

→ E-version in OUP Exercise Platform is also available.

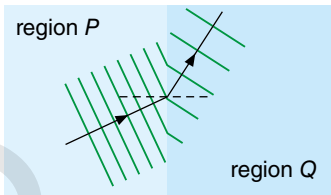
Google Forms Microsoft Forms

1 A straight wave travels from regions P to Q as shown. Determine whether the following statements are true.

(a) Region Q is deeper than region P. (T/F)

(b) The wave frequency remains unchanged. (T/F)

(c) A straight wave always changes its travelling direction when it travels from regions P to Q. (T/F)

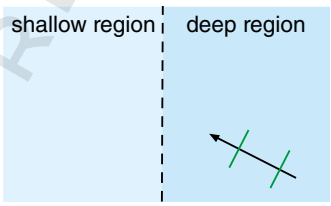


Learn by practice 2

1 A straight wave travels from a deep region to a shallow region as shown.

(a) Sketch the refracted wave in the shallow region.

(b) Label the angle of incidence i and angle of refraction r.



c Snell's law

Cross-link
Refer to Book 3A Ch 2.

Similar to the case of light, the refraction of a wave also obeys Snell's law, i.e. the ratio $\frac{\sin i}{\sin r}$ is a constant. In fact, the ratio of speeds and the ratio of wavelengths in the two media also equal to the same constant.

Consider a wave travelling from region 1 with wave speed v_1, wavelength λ_1 and angle of incidence θ_1 to region 2 with wave speed v_2, wavelength λ_2 and angle of refraction θ_2 (Fig 5.2i).

This equation can be applied in the refraction of light.

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

We may define the refractive index from region 1 to region 2 as $n_{1\to 2} = \frac{v_1}{v_2}$

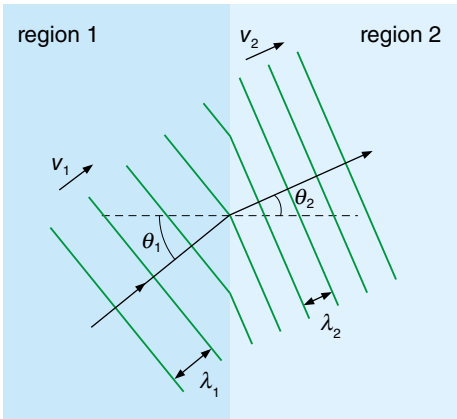


Fig 5.2i *A wave travelling from region 1 to region 2.*

In the discussion of refraction of light, we define the refractive index of a medium by comparing the speed of light in this medium to that in a VACUUM. This value is not affected by the travelling direction of the light ray. However, the discussion here involves two arbitrary media. We have to state clearly the travelling direction of the wave.

Know more

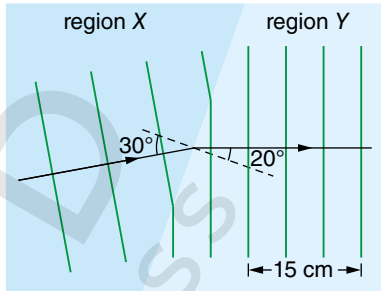
Derivation of Snell's law

We can derive Snell's law using geometry.

▸ p.100

Example 4 Snell's law

A straight water wave travels from region X to region Y as shown. The angle of incidence is 30° and the angle of refraction is 20°.



(a) Find the wavelength λ_X of the wave in region X.

(b) How will the angle of refraction change if the water depth in region Y is reduced?

Solution

(a) $\lambda_Y = \dfrac{15}{3} = 5$ cm

By $\dfrac{\sin\theta_1}{\sin\theta_2} = \dfrac{\lambda_1}{\lambda_2}$,

$\dfrac{\sin 30°}{\sin 20°} = \dfrac{\lambda_X}{5}$

$\lambda_X = 7.31$ cm

(b) Decrease

Tip

By $\dfrac{\sin\theta_X}{\sin\theta_Y} = \dfrac{v_X}{v_Y}$,

water depth in Y ↓

$\Rightarrow v_Y$ ↓ $\Rightarrow \sin\theta_Y$ ↓ $\Rightarrow \theta_Y$ ↓

Google Forms Microsoft Forms

Quick check 6

→ E-version in OUP Exercise Platform is also available.

1 A wave travels from region X to region Y. Complete the table.

	θ_X	θ_Y	λ_X	λ_Y	v_X / m s^{-1}	v_Y / m s^{-1}
(a)	50°	30°	-	-	0.5	
(b)	40°		8 cm	10 cm	-	-

Learn by practice 3

1 A wave of wavelength 10 cm crosses a boundary with an angle of incidence of 30° and an angle of refraction of 40°. Find the wavelength after crossing the boundary.

2 A wave travels at 4 m s^{-1} in region X and enters region Y with an angle of incidence of 60°. If its speed in region Y is 2 m s^{-1}, find the angle of refraction.

3 A water wave travels from region X to region Y. The angle of refraction is 70°. The wave speed in Y is double that in X.

(a) When the wave enters region Y,

(i) how does the frequency change?

(ii) how does the wavelength change?

(b) What is the angle of incidence?

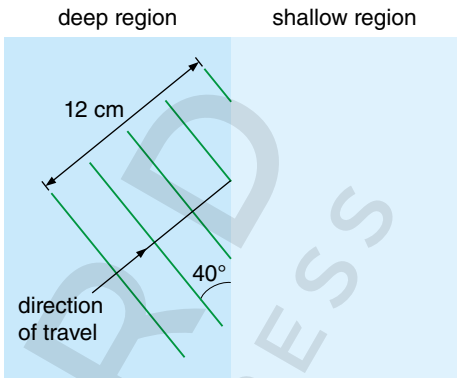
☆ HKDSE 2019 Paper 1B Q5

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Refraction of a water wave

A water wave of frequency 20 Hz travels from a deep region to a shallow region. Its wavelength in the shallow region is half that in the deep region. The wave pattern in the deep region when viewed from above is as shown.



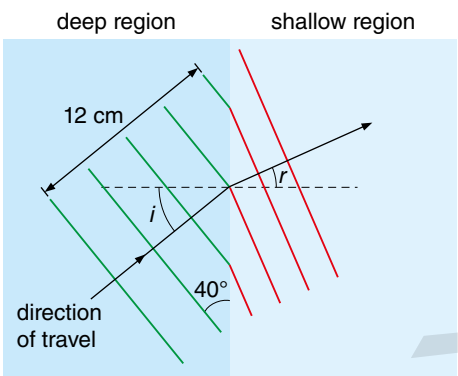
(a) Sketch the wave pattern in the shallow region. (2 marks)

(b) Find the wave speed in the shallow region. (2 marks)

(c) Find the angle of refraction of the wave. (2 marks)

Solution

(a)



(Shorter wavelength) 1A

(Bending towards normal) 1A

Common mistake
Students may wrongly draw the refracted wave as bending away from the normal.

Some may draw wavefronts that are not continuing across the boundary.

(b) Wavelength in shallow region $= \lambda_D \times \frac{1}{2}$

$= \frac{12}{4} \times \frac{1}{2}$

$= 1.5$ cm 1M

Common mistake
Students may not be aware that f remains unchanged.

Wave speed v in shallow region $= f\lambda$

$= 20 \times 1.5$

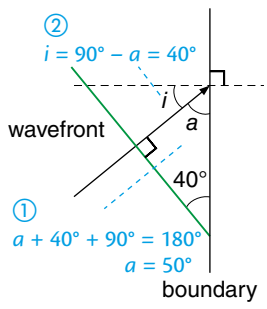
$= 30$ cm s^{-1} 1A

Common mistake
Students may not know that the angle of incidence is 40°.

(c) By $\frac{\sin\theta_1}{\sin\theta_2} = \frac{\lambda_1}{\lambda_2}$,

$\frac{\sin 40°}{\sin r} = \frac{2}{1}$ 1M

$r = 18.7°$ 1A



▶ Revision 5 Q32 (p.98)

Exercise 5.2

★**1** A straight wave propagates towards a straight barrier (Fig a).

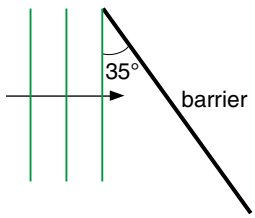
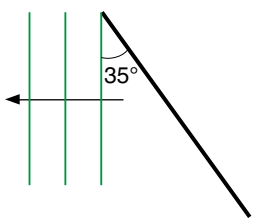


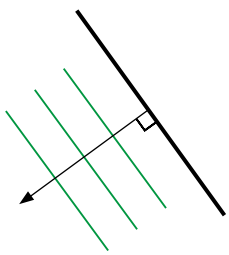
Fig a

Which of the following diagrams correctly shows the reflected wave?

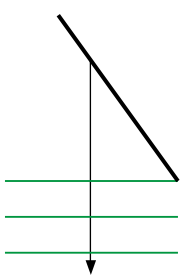
A



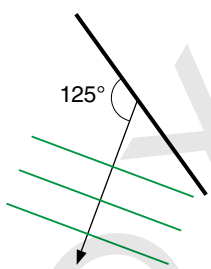
B



C



D



★**2** Figure b shows two straight waves, *P* and *Q*, produced in two ripple tanks *X* and *Y* respectively with the bars vibrating at the same frequency.

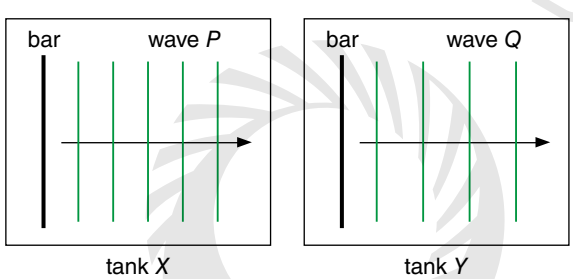


Fig b

Which of the following statements are correct?

(1) *P* travels at a constant speed.

(2) *P* and *Q* travel with the same speed.

(3) Tank *Y* has a larger water depth than tank *X*.

A (1) and (2) only B (1) and (3) only

C (2) and (3) only D (1), (2) and (3)

★**3** As a water wave travels from region *P* to region *Q*, its wave speed increases from 3 cm s^{-1} to 4 cm s^{-1}. The wavelength of the wave in region *Q* is 5 cm. Which of the following statements is/are correct?

(1) Region *Q* is deeper.

(2) The wavelength of the wave in region *P* is 10 cm.

(3) The frequency of the wave in region *P* is 0.8 Hz.

A (2) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

★**4** A dipper vibrates to produce a circular wave in a ripple tank. After 2 periods, the wavefronts shown in Figure c are observed. Which of the following best represents the wavefronts observed after another 2 periods?

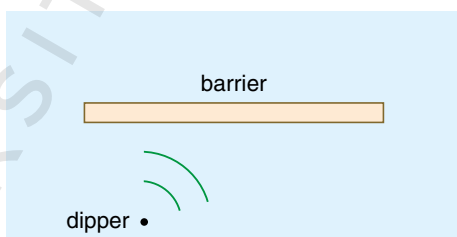
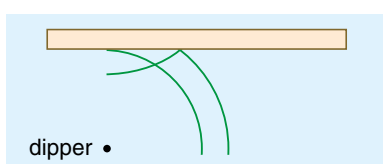
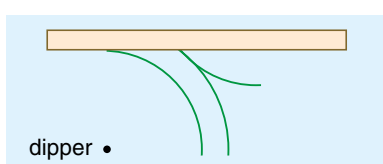


Fig c

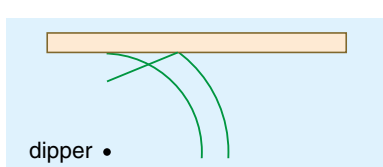
A



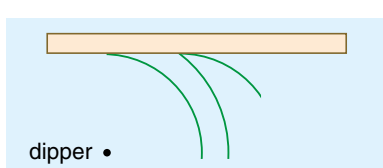
B



C



D



★**5** A ripple tank has two regions, *P* and *Q*, which are different in depth. A straight wave travels from *Q* to *P* (Fig d).

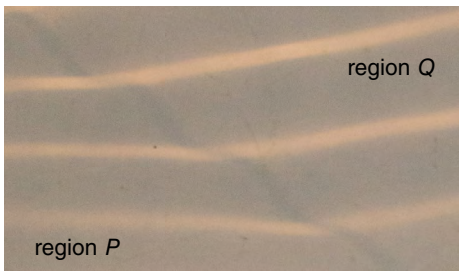


Fig d

(a) Mark the boundary between *P* and *Q* in Figure d.

(b) Name the phenomenon which occurs at the boundary.

(c) The speed of the water wave is 4 cm s^{-1} in *P* and 6 cm s^{-1} in *Q*. If the angle of incidence in *Q* is 60°, what is the angle of refraction in *P*?

★**6** A straight wave travels towards a straight barrier as shown (Fig e).

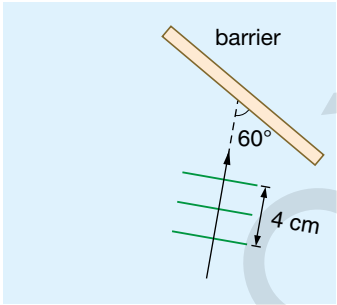


Fig e

(a) What is the wavelength of the wave?

(b) What is the angle of reflection?

(c) Sketch the reflected wavefronts in Figure e.

★**7** Figure f shows a water wave travelling from a deep region to a shallow region. In the deep region, the frequency of the wave is 2 Hz, and the wave speed is 6 cm s^{-1}.

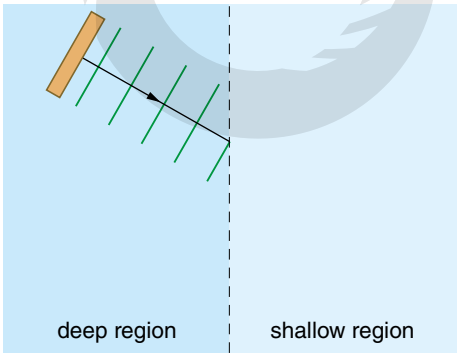


Fig f

(a) Sketch the wave pattern in the shallow region.

(b) Find the wavelength in the deep region.

(c) The ratio of the wavelengths in the two regions is 2 : 1. Find the angle of incidence in the deep region if the angle of refraction in the shallow region is 20°.

★**8** A straight water wave travels from region *P* to region *Q* as shown in Figure g. The wave speed in *P* and *Q* are 2 cm s^{-1} and 3 cm s^{-1} respectively.

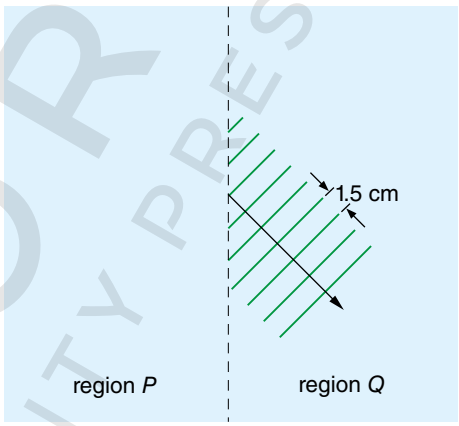


Fig g

(a) Which region is deeper?

(b) What is the wavelength of the wave in *P*?

(c) Sketch the wavefronts in *P*.

★**9**

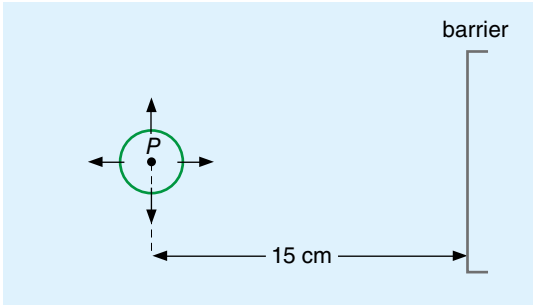


Fig h

Figure h shows a ripple tank with a straight barrier. A point dipper touches the water surface at *P* at time $t = 0$. The pulse produced reaches the barrier at $t = 0.6$ s.

(a) Find the speed of the pulse.

(b) When will the pulse return to *P*?

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5.3 Diffraction and interference

1 Diffraction

DSE goal

Describe diffraction of waves through narrow gap and around corner.

Examine effect of width of slit on degree of diffraction.

Draw wavefront diagrams to show diffraction.

DSE exam

14(1A)Q16, 17(1A)Q17

Mary is riding a pedal boat at position *P* in the *Wan Chai promenade* (Fig 5.3a). A straight wave produced by a big ship travels towards the breakwater. Although position *P* is behind the breakwater, Mary's boat is still affected by the wave! Do you know why?

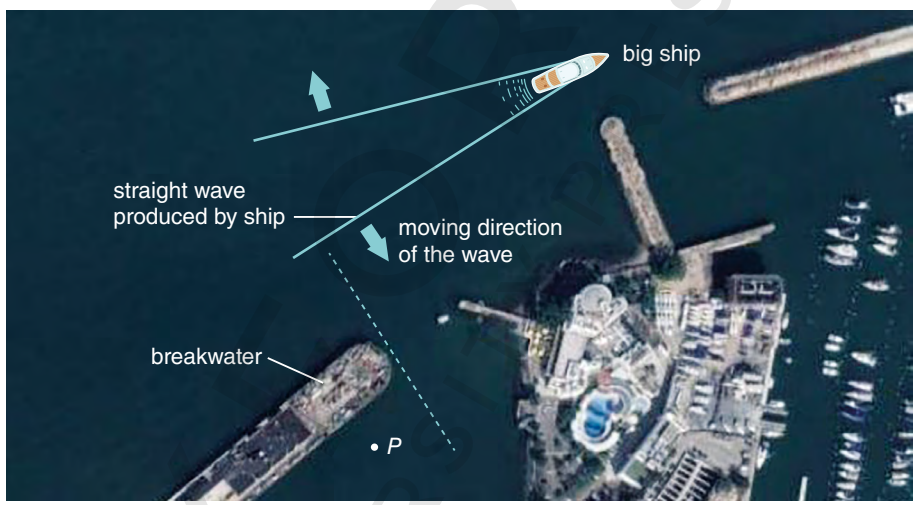


Fig 5.3a *Wan Chai promenade.*

a Spreading of waves

→ This simulation shows the diffraction of water waves at an edge, a gap and an obstacle. It is available on OUP web in PC version (.exe) only.

Simulation Video

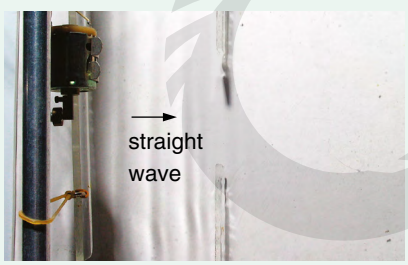
→ This video demonstrates Expt 5d.

Experiment 5d / Diffraction of water waves

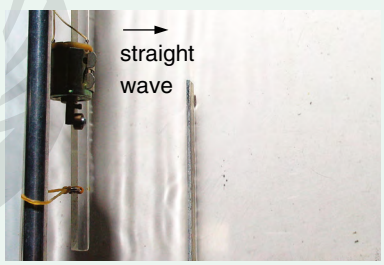
1 Set up a ripple tank.

2 Send a straight wave towards a gap. Observe the wave pattern after the wave passes through the gap. Change the gap width and the wavelength and see how the wave pattern changes.

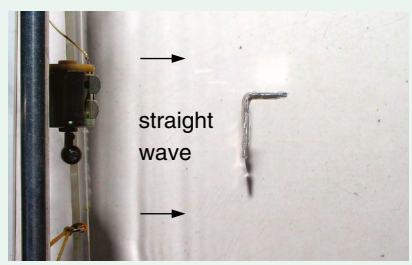
3 Repeat step 2 by sending the wave towards a straight barrier and then towards a small obstacle.



gap between two barriers



straight barrier



small obstacle

Discussion

How do the gap width and the wavelength affect the wave pattern?

Wan Chai promenade 灣仔海濱長廊

Table 5.3a shows what happens to the straight water wave when it passes through different settings in Experiment 5d.

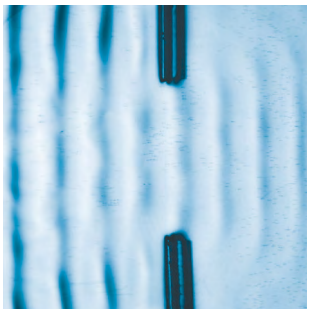
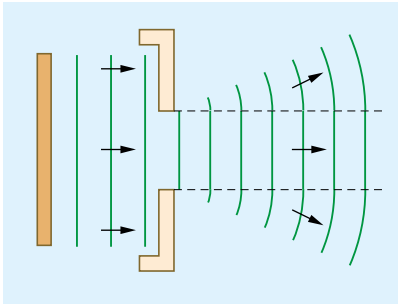
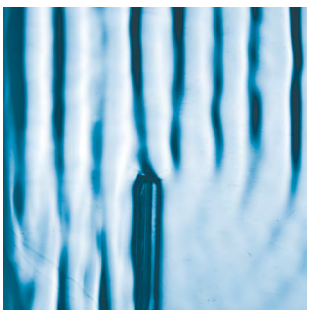
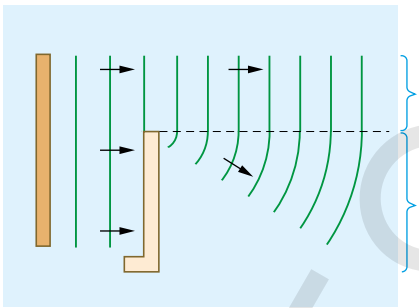
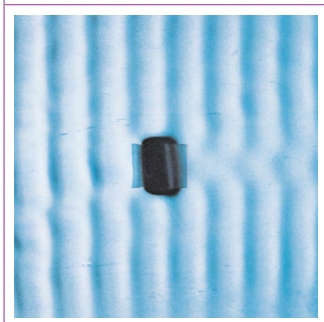
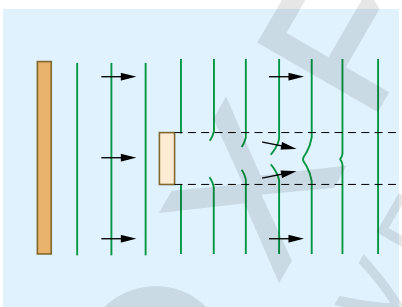
		The wave bends into the 'shadows' of the barriers and spreads out.
		The wave bends around the edge of the barrier into the 'shadow'.
		The wave bends around the edges of the obstacle. It closes up on the other side if the obstacle is small enough.

Table 5.3a *Straight waves bending around the edges of different barriers.*

The phenomenon shown in the experiment is called **diffraction**. Diffraction only causes the direction a wave travels to change. The wave speed, frequency and wavelength remain unchanged.

Diffraction is the bending or spreading of waves around the edge into the 'shadow' of an obstacle.

Diffraction is common in daily life. For example, the phenomenon occurs when a sea wave hits a breakwater (Fig 5.3b).

Diffraction explains why Mary's boat mentioned on p.58 is affected by the wave even though it is behind a breakwater.

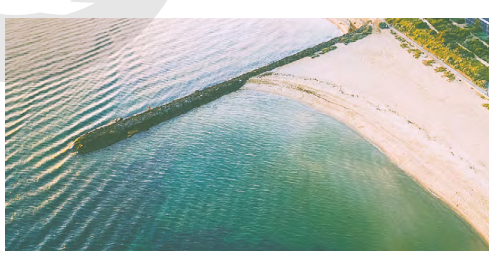




Fig 5.3b *Diffraction of sea waves behind breakwaters.*

diffraction 衍射 / 繞射

Simulation

→ This simulation shows the diffraction of water waves around a barrier and through a gap. Ss can vary the wavelength and the size of the gap.

b Factors affecting the degree of diffraction

In Experiment 5d, we can see that the wave spreads more when the wavelength λ increases (Fig 5.3c).

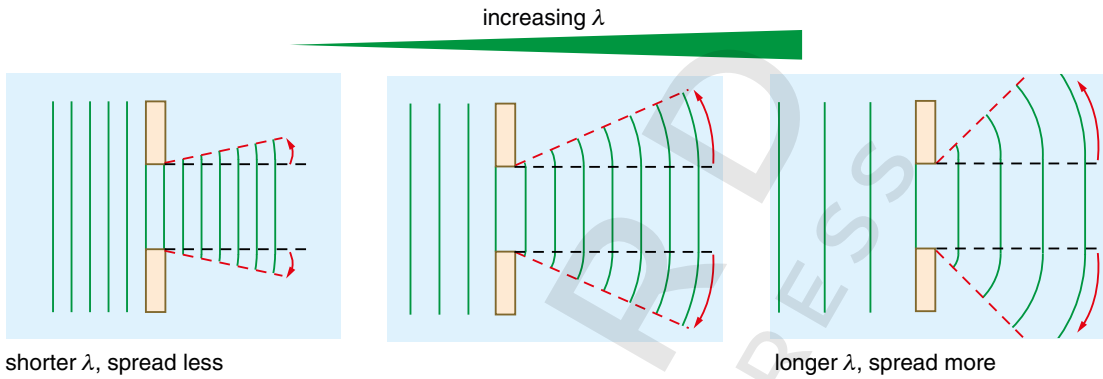


Fig 5.3c *Effect of λ on diffraction patterns.*

When the gap width decreases, the wave spreads more (Fig 5.3d).

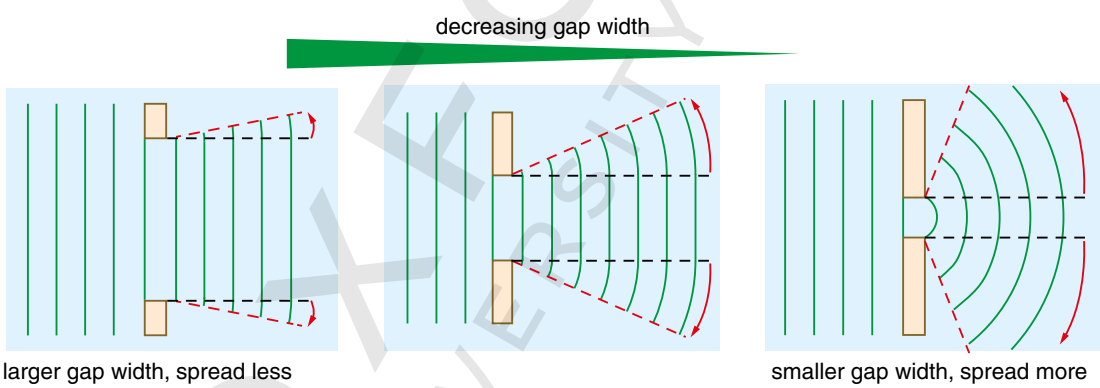


Fig 5.3d *Effect of gap width on diffraction patterns.*

When the gap width is about the same size as λ, the degree of diffraction is large.

When the gap is much smaller than λ, the wave cannot pass through the gap.

When a wave passes through a gap, it spreads more when
1 the wavelength increases;
2 the gap width decreases.

Google Forms | Microsoft Forms

Quick check 7

→ E-version in OUP Exercise Platform is also available.

1 A straight water wave is produced in a ripple tank. It passes through a gap and spreads into the 'shadow' of the obstacle. Determine whether the following statements are true.

(a) The wave frequency remains unchanged in the process. (T/F)

(b) If the wavelength increases, the degree of diffraction increases. (T/F)

(c) If the gap width increases, the degree of diffraction increases. (T/F)

(d) If the frequency increases without changing the water depth, the degree of diffraction decreases. (T/F)

(e) If the water depth increases without changing the frequency of the wave, the degree of diffraction increases. (T/F)

2 Interference

a Superposition of two pulses

DSE goal
Describe the superposition of two pulses.

When two cars collide, they may come to a stop. How about waves (Fig 5.3e)

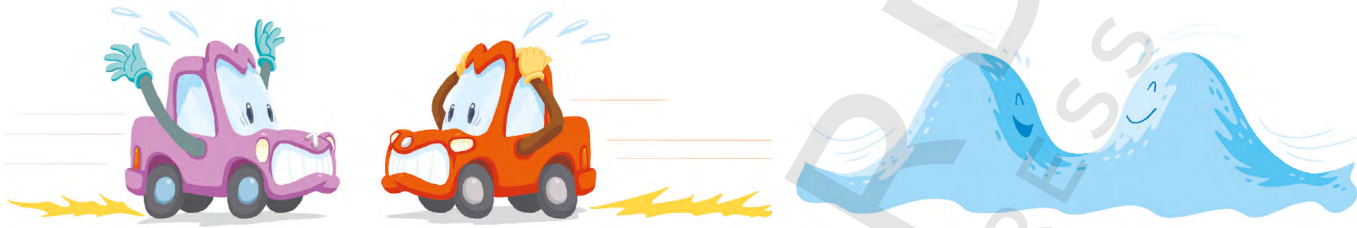


Fig 5.3e *What will happen when two waves meet?*

Let us find out the answer in Experiment 5e.

Simulation Video

→ This simulation shows what happens when two transverse pulses travelling in opposite directions meet and pass through each other. Ss can change the amplitude and shape of the pulses and observe its effect on the resultant pulse.

→ This video demonstrates Expt 5e.

✎ Emphasize that the comparison is made with the larger one among the two original pulses. Therefore, 'a smaller resultant pulse' may still be larger than one of the original pulses.

→ This is a Video Quiz about this experiment.

Experiment 5e / Adding pulses

✎ Record the result with a smartphone using slow motion mode. Observe the result again with the video.

1 Stretch a long spring on the floor.

2 Generate two pulses by flicking the two ends of the spring sideways in the same direction at the same time (Fig a). Observe what happens when the two pulses meet.

3 Repeat by generating pulses in opposite directions (Fig b).

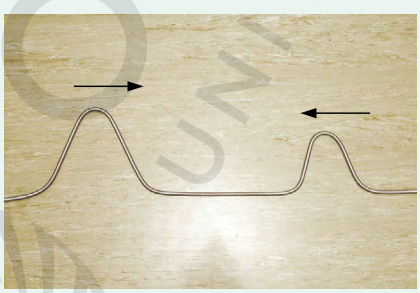


Fig a

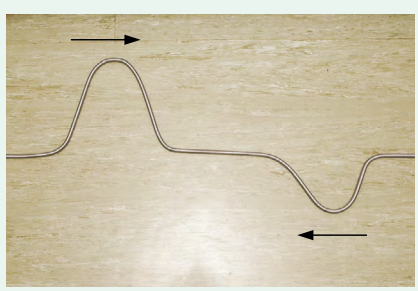


Fig b

Discussion

1 What happens when two pulses on the same side meet?

2 What happens when two pulses on opposite sides meet?

In Experiment 5e, if two pulses on the same side meet, the resultant pulse is larger; if two pulses on opposite sides meet, the resultant pulse is smaller (Fig 5.3f and 5.3g on p.62). After passing each other, the pulses keep their original shapes, speeds and directions of travel.

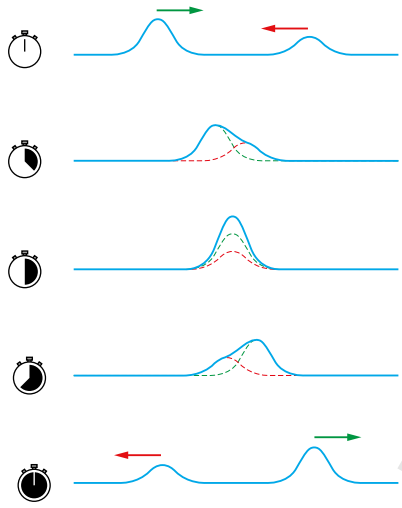


Fig 5.3f Two pulses on the same side meet.

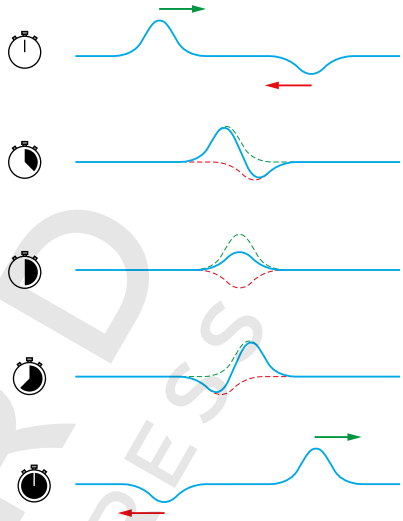
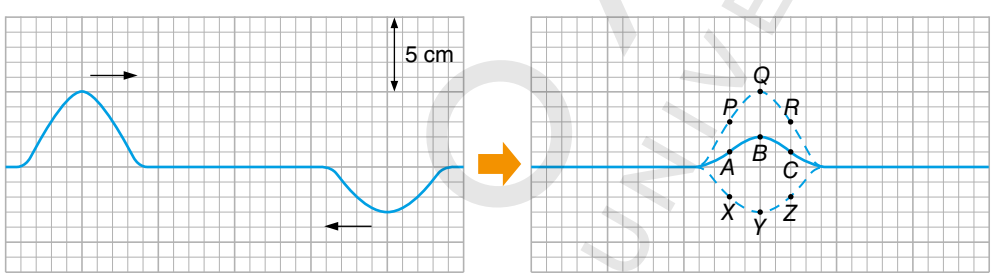


Fig 5.3g Two pulses on opposite sides meet.

We say that **superposition** occurs when two pulses or waves overlap. The **principle of superposition** states that (Fig 5.3h):

When two or more waves overlap, the displacement of any point on the resultant wave is the sum of the displacements due to all waves at that point.



Displacements s of A, B and C:
$s_A = s_P + s_X = 3 + (-2) = 1$ cm
$s_B = s_Q + s_Y = 5 + (-3) = 2$ cm
$s_C = s_R + s_Z = 3 + (-2) = 1$ cm

Fig 5.3h Applying the principle of superposition to find the displacements of particles on a resultant pulse.

Quick check 8

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

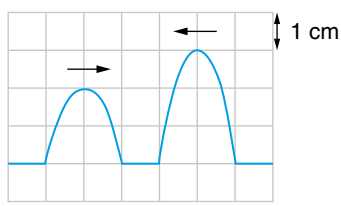
1 *True or false:*

(a) When two pulses meet, the resultant pulse must be larger than the original pulses. (T/F)

(b) A pulse will slow down after passing through another pulse. (T/F)

(c) A pulse will be reflected if it meets another pulse. (T/F)

2 Two pulses meet as shown. What is the maximum displacement of the resultant pulse?



superposition 疊加

DSE goal

Realize the interference of waves.

Examine the interference of waves from two coherent sources.

Draw wavefront diagrams to show interference.

b Interference of waves

When two waves overlap, **interference** occurs. Let us study the interference of waves in Experiment 5f.

✎ Ts may first display the wave pattern produced by one dipper for comparison.

Experiment 5f / Producing an interference pattern

1 Set up a ripple tank.

2 Attach two dippers to a vibrating bar to produce two circular waves. Observe the pattern produced.



Discussion

Describe the interference pattern. Are there points that always stay calm? Are there points that vibrate violently?

Simulation Video

→ This simulation shows different kinds of interference patterns formed by water waves. It is available on OUP web in PC version (.exe) only.

→ This video demonstrates Expt 5f.

In Experiment 5f, the two dippers are connected to the same vibrating bar. They have the **same frequency and a constant phase difference** (constantly in phase in this case). We say that the two dippers are **coherent sources**. When the waves produced by **two coherent sources** overlap, a **steady interference pattern** is formed (Fig 5.3i).

▶ Two sources vibrating in phase must be coherent.

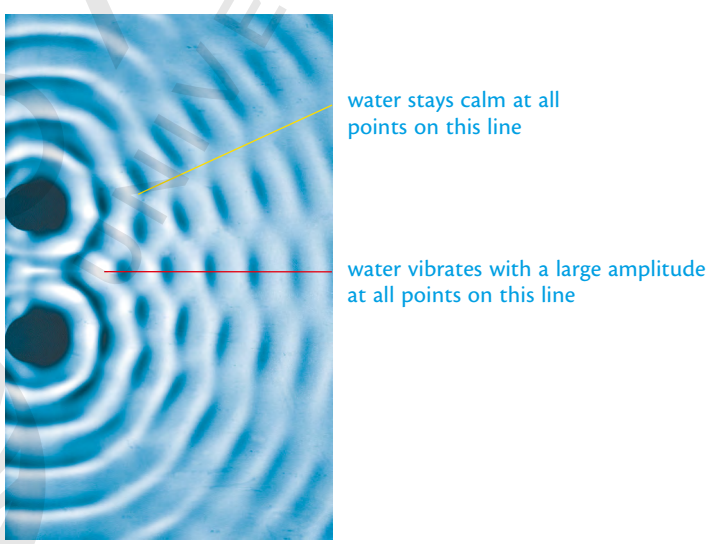


Fig 5.3i *Interference pattern formed by two circular waves.*

The following can be observed from the pattern.

1 Some parts stay calm all the time. They do not move at all.

2 Some parts vibrate with a larger amplitude than when there is only one wave. **Outgoing** bright and dark arcs can be seen at these parts. The bright arcs are the crests and the dark arcs are the troughs.

interference 干涉　　coherent sources 相干源

DSE goal

Distinguish between constructive and destructive interference.

Determine conditions for constructive and destructive interference in terms of path difference.

DSE exam

15(1A)Q13, 17(1B)Q6(b), 21(1B)Q14, 22(1A)Q19

Simulation

→ This simulation shows how two waves add up to produce a resultant wave. Ss can change the phase difference and observe how constructive and destructive interference are produced.

C Understanding the interference pattern

i Using phase difference

Consider two sources S_1 and S_2 vibrating in phase and producing waves of the same amplitude. Figure 5.3j shows the waves produced **at a certain time instant** t_1. The purple and green lines represent the crests of waves from S_1 and S_2 respectively.

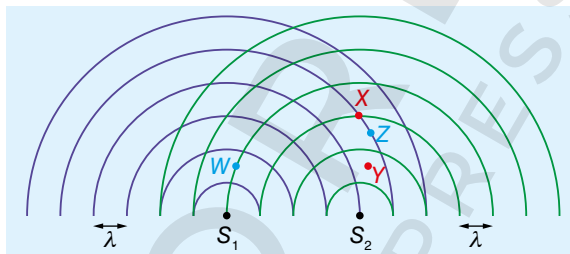


Fig 5.3j *Wavefronts produced by S_1 overlap with those produced by S_2 at $t = t_1$.*

At point X, a crest from S_1 meets a crest from S_2 (Fig 5.3k). The crests add up to form a higher crest (Fig 5.3l). Similarly, at point Y, a trough from S_1 meets a trough from S_2 (Fig 5.3m). They add up to form a deeper trough (Fig 5.3n). At both points, the two waves are **always in phase** and reinforce each other, giving rise to **constructive interference**.

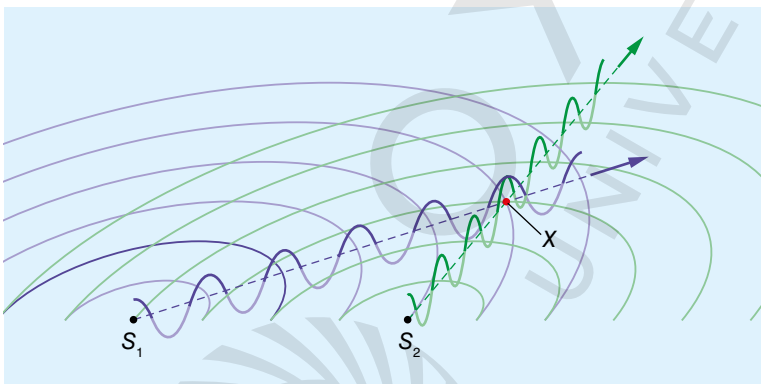


Fig 5.3k *At point X, a crest from S_1 meets a crest from S_2.*

✎ Emphasize that the displacement of the particle at X (or Y) are changing continuously. At some instants, the two waves arrive at X with zero displacement and the resultant displacement is thus zero.

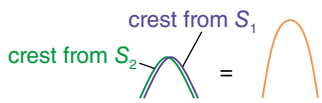


Fig 5.3l *The crests add up to form a higher crest.*

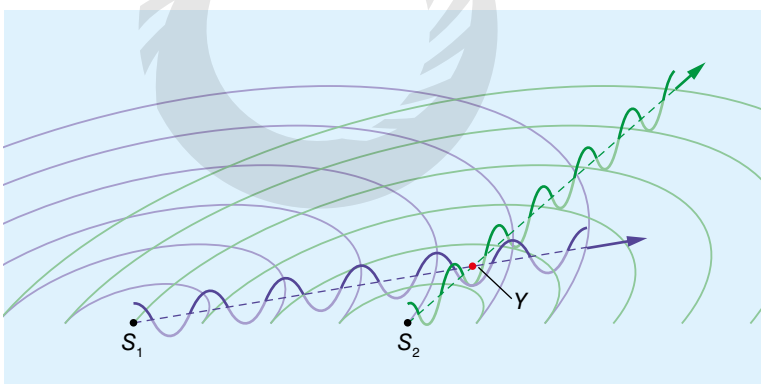


Fig 5.3m *At point Y, a trough from S_1 meets a trough from S_2.*

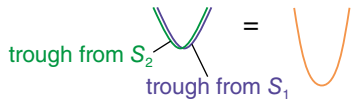


Fig 5.3n *The troughs add up to form a deeper trough.*

constructive interference 相長干涉

Two waves cancel each other out entirely only if they have the same amplitude. Unless otherwise specified, two waves that interfere with each other are assumed to have the same amplitude in this book.

At point Z (Fig 5.3j on p.64), a crest from S_1 meets a trough from S_2 (Fig 5.3o). They cancel each other out and the particle there is at the equilibrium position (Fig 5.3p). A similar situation happens at point W. At both points, the two waves are **always in antiphase**, giving rise to **destructive interference**.

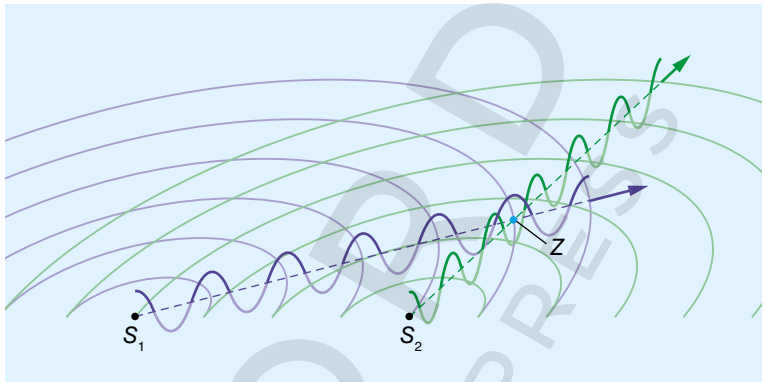


Fig 5.3o *At point Z, a crest from S_1 meets a trough from S_2.*

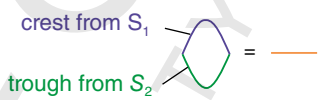


Fig 5.3p *The crest and the trough cancel out each other.*

A particle at a position of constructive interference does not always stay at a crest or a trough. As the waves travel, the particle moves from a crest to a trough and then back to a crest again. Therefore, the particle **keeps vibrating** with an amplitude larger than when only one wave is present. Figure 5.3q shows the displacement–time graph of the particle at X.

By contrast, a particle at a position of destructive interference **always remains stationary at the equilibrium position**. Figure 5.3r shows the displacement–time graph of the particle at Z.

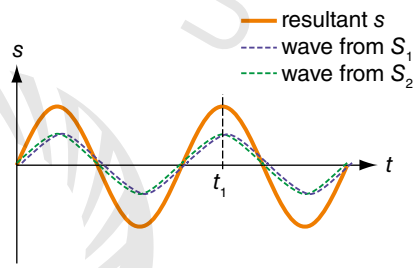


Fig 5.3q *s–t graph of the particle at X.*

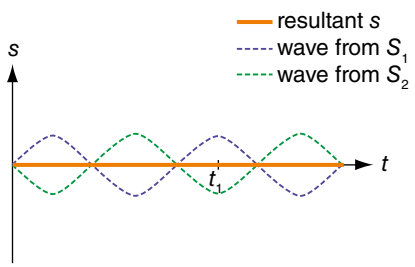


Fig 5.3r *s–t graph of the particle at Z.*

- **Constructive interference occurs at positions where two waves arrive in phase. Particles at those positions vibrate with a larger amplitude.**
- **Destructive interference occurs at positions where two waves arrive in antiphase. Particles at those positions remain stationary.**

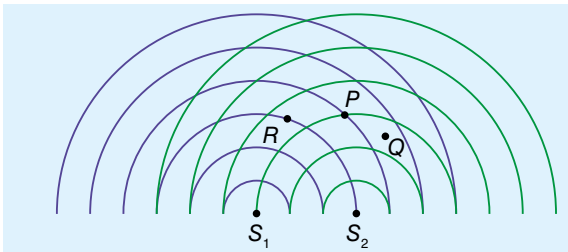
destructive interference 相消干涉

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 The figure shows the waves from two sources S_1 and S_2 vibrating in phase. The lines represent the crests of the waves. Determine whether the following statements are true.



(a) Constructive interference occurs at P. (T/F)

(b) Destructive interference occurs at Q. (T/F)

(c) The particle at Q always stays at a trough. (T/F)

(d) The particle at R remains stationary. (T/F)

(e) If the displacement of a point is zero at a certain instant, destructive interference must occur at this point. (T/F)

(f) Constructive interference only occurs when a crest meets a crest or a trough meets a trough. (T/F)

ii Using path difference

Besides the phase difference, we can also use the **path difference** (Δ) to determine whether constructive or destructive interference occurs at a certain point. The path difference at a point is the difference between the distances of the point from the two sources. For example, in Figure 5.3s, the path difference at X is $S_1X - S_2X$.

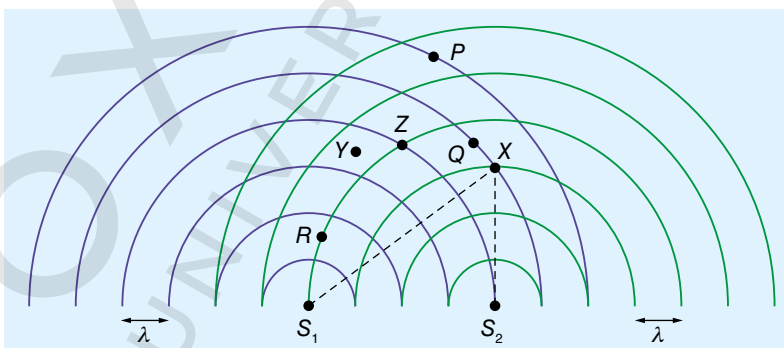


Fig 5.3s *Path difference at $X = S_1X - S_2X$.*

Point sources S_1 and S_2 vibrate in phase. The types of interference and path differences at different positions are tabulated below (Table 5.3a).

	Type of interference	Distance from S_1	Distance from S_2	Path difference (Δ)
X	constructive	5λ	3λ	2λ
Y	constructive	3.5λ	4.5λ	1λ
Z	constructive	4λ	4λ	0λ
P	destructive	6λ	5.5λ	0.5λ
Q	destructive	5λ	3.5λ	1.5λ
R	destructive	1.5λ	4λ	2.5λ

Table 5.3a *Type of interference and path difference.*

May ask Ss to fill in the type of interference and the path difference for various points on Fig 5.3s and ask Ss to generalize the relationship by themselves.

path difference 程差

Looking at Table 5.3a, can you observe the relationship between the type of interference and the path difference?

For two coherent point sources vibrating in phase:

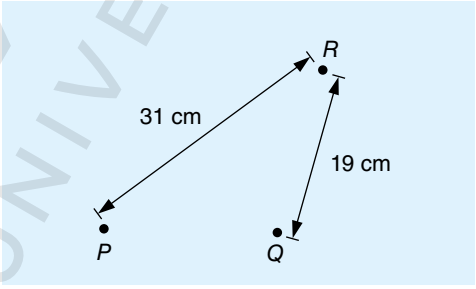
- **Constructive interference occurs at positions where the path difference is equal to whole numbers of wavelengths (0, λ, 2λ…)**
- **Destructive interference occurs at positions where the path difference is equal to 0.5λ, 1.5λ, 2.5λ…**

It is equivalent to understand interference using phase difference and path difference. For two sources vibrating in phase, when the path difference at a certain point from the two sources is $n\lambda$, the waves arrive there in phase. When the path difference at a certain point is $(n + 0.5)\lambda$, the waves arrive there in antiphase.

n = 0, 1, 2...

Example 5 Determining the type of interference

The figure shows two dippers placed at P and Q. They vibrate in phase to produce circular water waves of wavelength 4 cm in a ripple tank.



(a) Determine the type of interference at R.

(b) If the wavelength of the wave is doubled, how will the type of interference at R be affected?

Solution

(a) Path difference at R = 31 − 19 = 12 cm = 3λ

$\therefore$ Constructive interference occurs at R.

(b) New wavelength = 2×4 = 8 cm

Path difference at R = 12 cm = 1.5λ

$\therefore$ Destructive interference will occur at R.

Google Forms Microsoft Forms

Quick check 10

→ E-version in OUP Exercise Platform is also available.

1 Two sources vibrate in phase and the waves produced overlap. Determine whether the following statements are true.

(a) If the path difference at a point is 0, constructive interference occurs at that point. (T/F)

(b) If the path difference at a point is λ, the displacement of the particle at that point will not be zero. (T/F)

(c) If the path difference at a point is 3λ, the amplitude of vibration at that point will not be zero. (T/F)

(d) If the path difference at a point is 2.5λ, the displacement of the particle at that point will always be zero. (T/F)

2 Two coherent dippers, *P* and *Q*, vibrate in phase and generate water waves in a ripple tank. The wavelength of the water waves is 4 cm. *X*, *Y* and *Z* are three points in the ripple tank. Which type of interference occurs at these points?

	Distance from *P*	Distance from *Q*	Type of interference
X	8 cm	12 cm	
Y	17 cm	9 cm	
Z	20 cm	14 cm	

d Nodal lines and antinodal lines

In the interference pattern formed in Experiment 5f, there are lines where the water is calm. They are called **nodal lines**. Destructive interference occurs on **every point** on these lines. There are also lines where the water vibrates with a large amplitude. They are called **antinodal lines**. Constructive interference occurs on **every point** on these lines (Fig 5.3t).

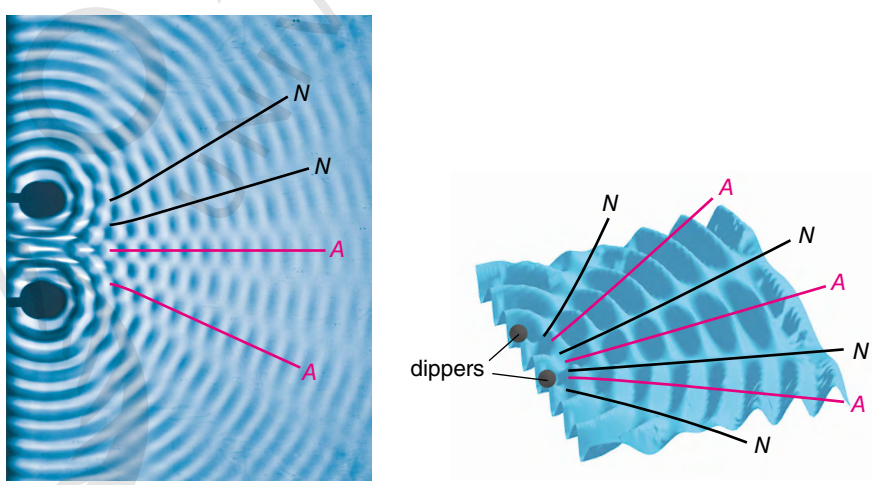


Fig 5.3t *Nodal lines (N) and antinodal lines (A).*

Note that:

1 The central antinodal line is always a straight line while all other lines are curves.

2 Bright and dark arcs (crests and troughs) travel outwards along the antinodal lines.

nodal line 節線　　antinodal line 腹線

Remind Ss that interference happens even if the two sources are not coherent. However, the interference pattern is not stable and thus hardly observable.

Every point on the same antinodal (or nodal) line has the same path difference. For example, in Figure 5.3u, the path difference at any point on the central antinodal lines A_0 is 0 and that on the nodal lines N_1 is 0.5λ.

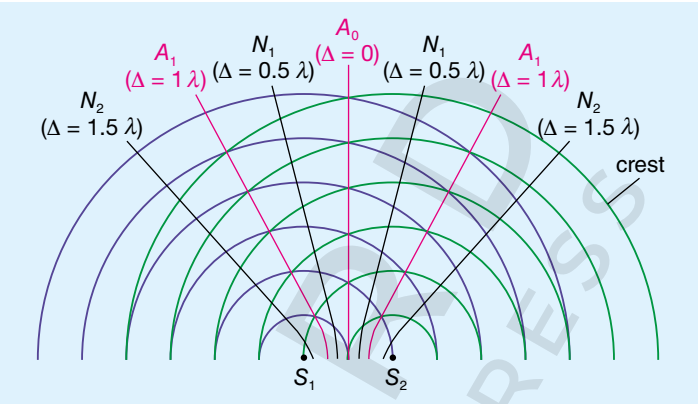


Fig 5.3u *Path difference of antinodal lines (A) and nodal lines (N).*

As the waves keep travelling outwards, the wavefronts are not stationary. However, if the sources are **coherent**, the locations of the nodal lines and antinodal lines will remain unchanged. This will result in a **steady** interference pattern. If the two sources are **incoherent** (i.e. not coherent), the two waves no longer have a constant phase difference at a point. The locations of the nodal lines and antinodal lines will change with time.

Watch the following videos to compare the wave patterns produced by coherent sources and incoherent sources.

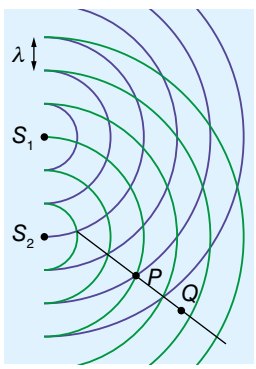
Video

→ The video 'Wave pattern produced by coherent source' demonstrates the wave pattern produced by a coherent source.

→ The video 'Wave pattern produced by incoherent source' demonstrates the wave pattern produced by a incoherent (i.e. not coherent) source.

Example 6 | Interference pattern

The figure shows water waves of wavelength λ produced by sources S_1 and S_2 vibrating in phase. The purple lines and green lines represent wave crests.



(a) Find the path differences at points P and Q in terms of λ.

(b) What type of interference occurs at Q?

(c) If S_1Q is 15 cm and S_2Q is 11 cm, find λ.

(d) Describe the motion of a particle at P at this instant.

Solution

(a) Path difference at Q = path difference at $P = 5\lambda - 3\lambda = 2\lambda$

(b) Constructive interference

(c) $\Delta = S_1Q - S_2Q$

$2\lambda = 15 - 11$

$\lambda = 2$ cm

(d) It is momentarily at rest.

Tip

P and Q lie on the same antinodal line, so they have the same Δ.

Tip

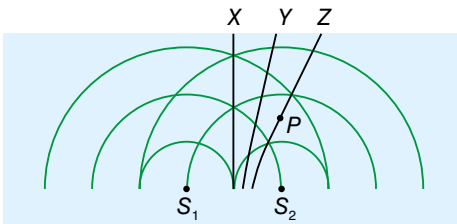
It is at the crest at this instant. It will then move downwards to the trough and then back to the crest in one period.

Google Forms | Microsoft Forms

Quick check 11

→ E-version in OUP Exercise Platform is also available.

1 The figure shows waves of wavelength λ produced by two sources S_1 and S_2 vibrating in phase. The green lines represent wave crests.



Determine whether the following statements are true.

(a) The path difference at any point on line X is λ. (T/F)

(b) The path difference at any point on line Y is 1.5λ. (T/F)

(c) A particle at P has zero displacement at the instant shown. (T/F)

2 Two water waves of different wavelengths overlap. At a certain instant, constructive interference occurs at point P. Determine whether the following statements are true.

(a) Constructive interference always occurs at P. (T/F)

(b) The principle of superposition can be applied at P to find the resultant displacement. (T/F)

DSE exam

12(1B)Q6(a)–(c), 19(1A)Q16

Simulation Video

→ This simulation shows an interference pattern when two sets of circular waves overlap. Ss can change the wavelength and the separation between the sources and observe the effects on the pattern.

→ This video demonstrates Expt 5g.

e Factors affecting an interference pattern

Experiment 5g Factors affecting an interference pattern

1 Set up a ripple tank. Attach two dippers to the same vibrating bar to produce an interference pattern.



2 See how the interference pattern changes when the following change.

(i) Wavelength (by changing the vibrating frequency of the dippers)

(ii) Separation between dippers

Discussion

1 What happens to the interference pattern when the wavelength of the waves increases?

2 What happens to the interference pattern when the separation between the dippers increases?

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In Experiment 5g, we can see that when the wavelength λ increases, the nodal and antinodal lines become more widely spaced (Fig 5.3v).

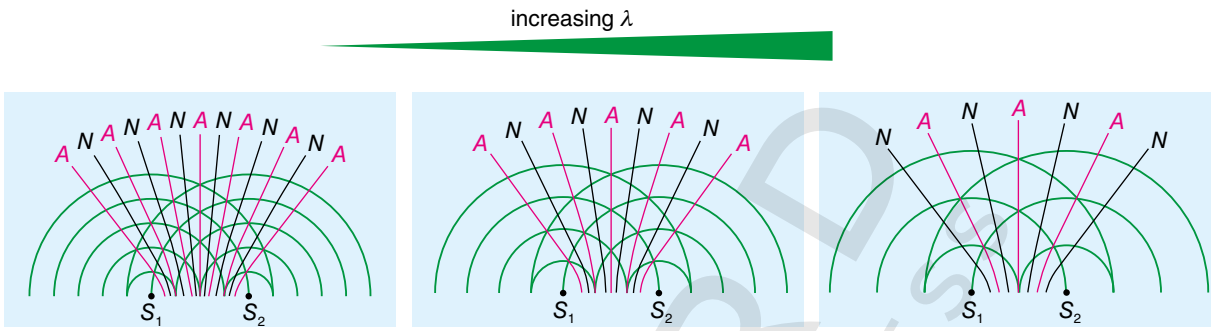


Fig 5.3v *Effect of λ on the interference pattern.*

If the separation is much larger than λ, the nodal lines will be very close together and the interference pattern cannot be observed.

When the separation D between the sources decreases, the nodal and antinodal lines become more widely spaced (Fig 5.3w).

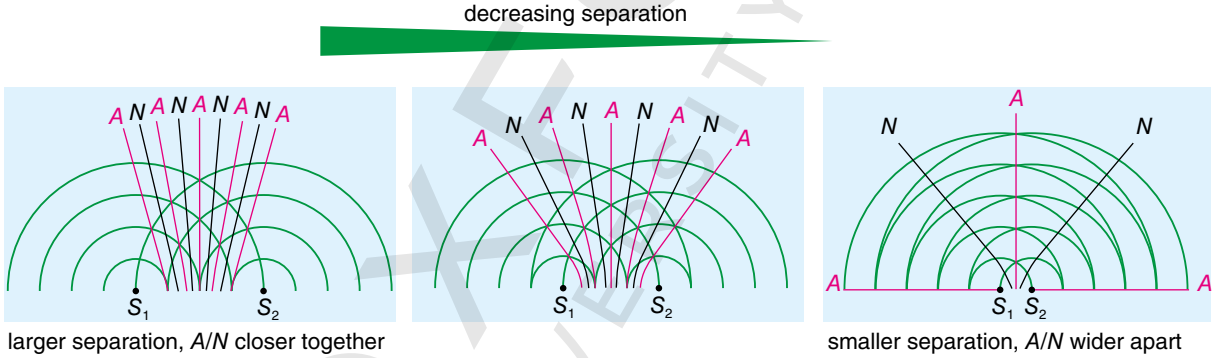


Fig 5.3w *Effect of the separation between the sources on the interference pattern.*

The nodal and antinodal lines in the interference pattern become more widely spaced when:

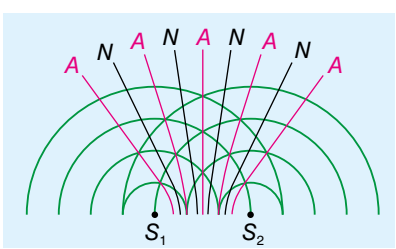
- **the wavelength of the waves increases;**
- **the separation between the sources decreases.**

Quick check 12

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Two dippers connected to the same vibrating bar are used to produce two circular waves. Which of the following changes will increase the separation between the antinodal lines?



- ☐ Increasing the wavelength of the water wave.
- ☐ Increasing the separation between the two sources.
- ☐ Increasing the frequency of the vibrating bar.
- ☐ Increasing the water depth in the ripple tank.

Example 7 Interference and vibrations

Two dippers S_1 and S_2 vibrate in phase and produce water waves of wavelength 2 cm (Fig a). The green lines represent wave crests. AB is the perpendicular bisector of S_1S_2. Along XY, maxima occur at some points and minima occur at some other points.

Water vibrates with the maximum amplitude at maxima; while water is calm at minima.

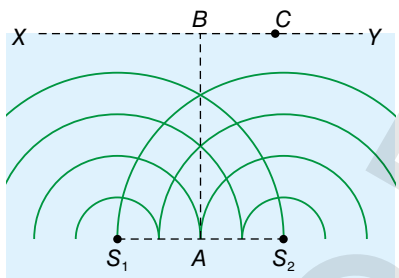


Fig a

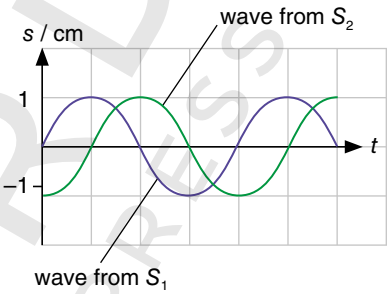


Fig b

(a) Explain why a maximum occurs at point B.

(b) Point C is the second minimum from point B. Find the path difference (in cm) at C.

(c) Sketch how the amplitude of the water wave varies along the line AB.

(d) Figure b shows the variation of the displacement s due to each wave with time t at a certain point on XY. Do the two waves arrive at this point in phase?

Two sources vibrating in antiphase are coherent.

(e) If S_1 and S_2 vibrate in antiphase, will a maximum or a minimum occur at B? Explain.

Solution

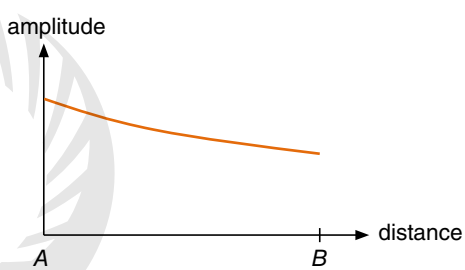
Tip
For 1st min, $\Delta = 0.5\lambda$
For 2nd min, $\Delta = 1.5\lambda$

(a) The path difference at B is zero, so constructive interference occurs.

(b) Path difference $\Delta = 1.5\lambda = 1.5(2) = 3$ cm

Tip
The amplitude of a circular wave decreases gradually with the distance from the source.

(c)



Tip
The crests of the two waves do not arrive at the same time.

(d) No

(e) The waves from S_1 and S_2 arrive B in antiphase.

$\therefore$ Destructive interference always occurs.

$\therefore$ A minimum will occur at B.

Tip
A maximum will become a minimum, and vice versa.

Practice 5.3 Q6 (p.75)

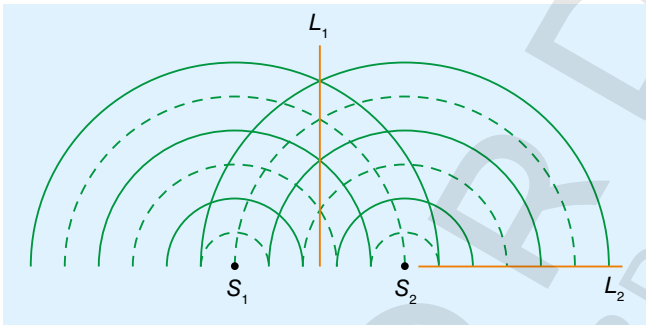
☆ HKDSE 2019 Paper 1A Q16, HKDSE 2012 Paper 1B Q6

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Nodal lines and antinodal lines

Water waves are produced by two sources S_1 and S_2 vibrating in phase. The figure shows the waves at a certain instant. The solid green lines represent wave crests and dotted green lines represent wave troughs.



(a) Draw a line X to connect points with path difference 1.5λ. (1 mark)

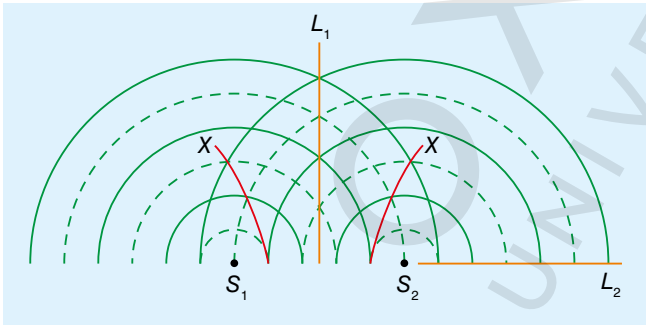
(b) How will the separation between L_1 and X change if S_1 and S_2 become wider apart? (1 mark)

(c) Is L_2 a nodal line or an antinodal line? Explain. (2 marks)

(d) Find the total number of antinodal lines formed. (2 marks)

Solution

(a)



Emphasize that a nodal line (or antinodal line) can be a straight line or a smooth curve. A suitable line is drawn through points with specific path difference. Ss may have difficulties in drawing a curve directly. Ts may ask them to mark the points before drawing the curve.

(Either one) 1A

Common mistake
Students may wrongly connect the points with a straight line. They do not know they should use a smooth curve.

(b) Decrease 1A

(c) For any point P on L_2,

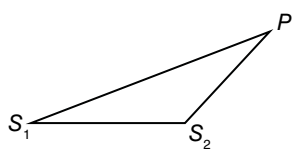
$\Delta = S_1P - S_2P = S_1S_2 = 2.5\lambda$ 1M

$\therefore$ L_2 is a nodal line. 1A

Common mistake
Students may not know that Δ cannot be larger than S_1S_2.

(d) $\Delta < S_1S_2 = 2.5\lambda$ 1M

In $\triangle S_1S_2P$,



$S_1S_2 + S_2P > S_1P$

$S_1S_2 > S_1P - S_2P = \Delta$

There is one line with $\Delta = 0$, two lines with $\Delta = \lambda$ and two lines with $\Delta = 2\lambda$.

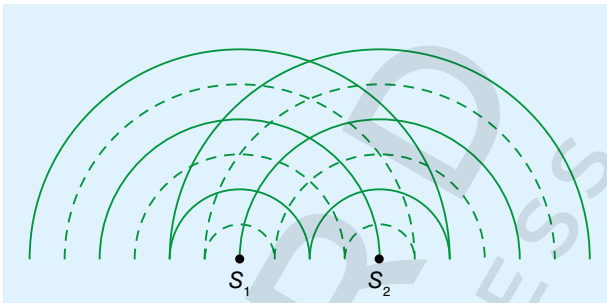
$\therefore$ The total number of antinodal lines is 5. 1A

Revision 5 Q18 (p.93), Q30 (p.97)

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Learn by practice 4

1 Dippers S_1 and S_2 vibrate in phase and water waves are produced as shown. Solid lines represent crests and dotted lines represent troughs.



(a) Draw and label a line L_1 to connect points with path difference 1λ.

(b) Draw and label a line L_2 to connect points with path difference 0.5λ.

f Diffraction and interference as the tests for waves

Apart from waves, other objects also exhibit reflection and refraction. For example, a ball bounces back when it hits a wall; a sledge changes direction when it travels from snow into mud at an angle. However, only waves exhibit diffraction and interference. Therefore, these two criteria can be used to determine whether something is a wave or not.

All waves exhibit reflection, refraction, diffraction and interference. Among these phenomena, only diffraction and interference are unique to waves.

Exercise 5.3

★**1** When a straight wave travels past an obstacle, it spreads into the shadow of the obstacle. Which of the following statements is/are correct?

(1) This phenomenon is called diffraction.

(2) The speed of the wave decreases as the wave spreads.

(3) The wavelength of the wave increases as the wave spreads.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★**2** Two sets of circular waves produced by coherent sources interfere with each other. Which of the following statements is/are correct?

(1) The positions of the wavefronts change with time.

(2) The positions where constructive interference occurs change with time.

(3) The displacement of the particle at a position where constructive interference occurs changes with time.

A (1) only B (1) and (2) only

C (1) and (3) only D (2) and (3) only

★**3** Two coherent dippers produce an interference pattern in a ripple tank (Fig a). Which of the following changes can cause the nodal and antinodal lines to be more closely spaced?



Fig a

(1) The frequency of the dippers increases.

(2) The depth of water in the tank decreases.

(3) The dippers are placed closer together.

A (1) only B (2) only

C (1) and (2) only D (2) and (3) only

★**4** A straight water wave spreads out when it travels past a gap (Fig b).

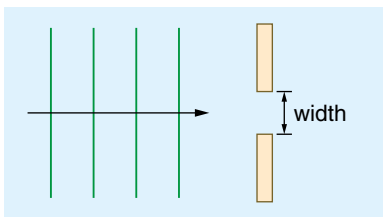


Fig b

Which of the following changes can make the wave spread out less?

(1) Increase the frequency of the wave.

(2) Increase the width of the gap.

(3) Increase the depth of the water.

A (1) only B (2) only

C (1) and (2) only D (1), (2) and (3)

★**5** Two pulses of the same shape, but on opposite sides, travel along a string towards each other as shown (Fig c).

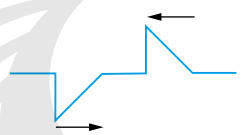


Fig c

Which of the following can be the resultant waveform when the pulses overlap with each other?

A

B 

C 

D 

★**6** Two sources, S_1 and S_2, vibrating in phase produce circular waves in a ripple tank. Figure d shows the interference pattern observed at a certain instant. Solid lines represent crests and dotted lines represent troughs.

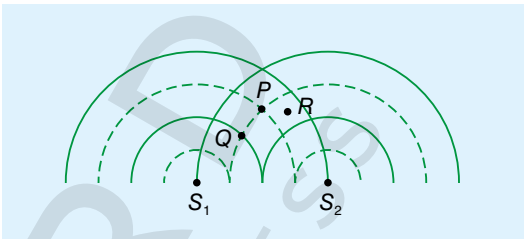


Fig d

(a) Determine the type of interference at P, Q and R.

(b) Describe the motion of the particles at P, Q and R at this instant.

(c) Sketch the displacement–time graphs for the particles at P, Q and R from this instant to a period later.

(d) Draw a line X to connect points with path difference 0.5λ. Name this line.

(e) Draw a line Y to connect points with path difference λ. Name this line.

(f) If S_1 and S_2 vibrate in antiphase, determine the type of interference at P.

★**7** Two vibrators, P and Q, vibrate in phase at frequency 15 Hz to generate water waves in a ripple tank (Fig e). The wave speed is 30 cm s^{-1}.

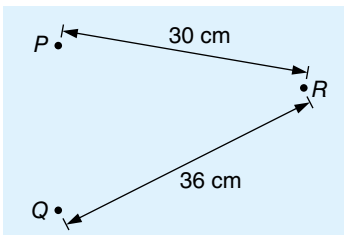


Fig e

(a) Find the wavelength of the waves.

(b) Do the waves from P and Q arrive at R in phase or in antiphase?

(c) Determine which type of interference takes place at R.

(d) If the frequency of the vibrators is halved, how will the answer in (c) change?

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5.4 Stationary waves

When you flick the free end of a long stretched string, a wave will travel along the string. What will happen if you flick a stretched string with both ends fixed, e.g. a guitar string (Fig 5.4a)? A wave will still be formed, but it will stay at the same position rather than travel along the string. Such kind of waves are called **stationary waves** or **standing waves**.



Fig 5.4a *Waves on guitar strings.*

→ This is a Video Quiz about this experiment.

Simulation Video

→ This simulation shows the reflection of waves at a fixed end. It also shows the formation of stationary waves. It is available on OUP web in PC version (.exe) only.

→ This video demonstrates Expt 5h.

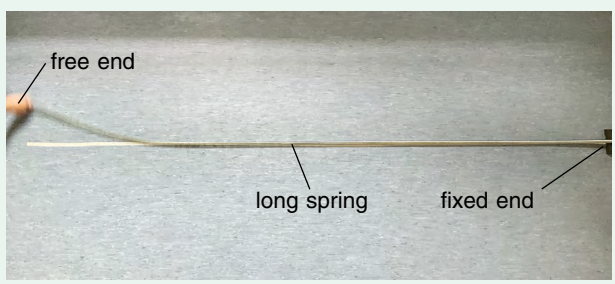
1 Observing stationary waves

Experiment 5h Stationary waves

1 Stretch a long spring on the floor, with one end fixed to a rigid support.

2 Flick the free end once to send a pulse along the spring. Observe how the pulse is reflected at the fixed end.

3 Flick the free end continuously. Slowly increase the frequency. Observe the wave pattern produced on the spring.



Record a video of the wave motion with a smartphone by using slow motion mode. Observe the result again with the video.

In the experiment, Ts may ask Ss to flick the spring with a smaller frequency so that the motion of the spring can be observed even without using slow motion video.

Discussion

1 What happens to the pulse when it arrives at the fixed end?

2 Describe the wave pattern produced in step 3.

stationary wave / standing wave 駐波

The wave appears as loops due to persistence of vision. This refers to the effect when vision seems to persist for a brief moment even after the light that causes the vision stops entering the eyes.

In Experiment 5h, when the spring is flicked continuously with a gradually increasing frequency, at certain stages one or more vibrating **loops** of large amplitudes are formed on the spring (Fig 5.4b). Snapshots of the spring show that the loops are actually a wave that does **not** move forwards or backwards. Such a wave is called a stationary wave.

Video

→ This video shows a stationary wave formed on a long spring.

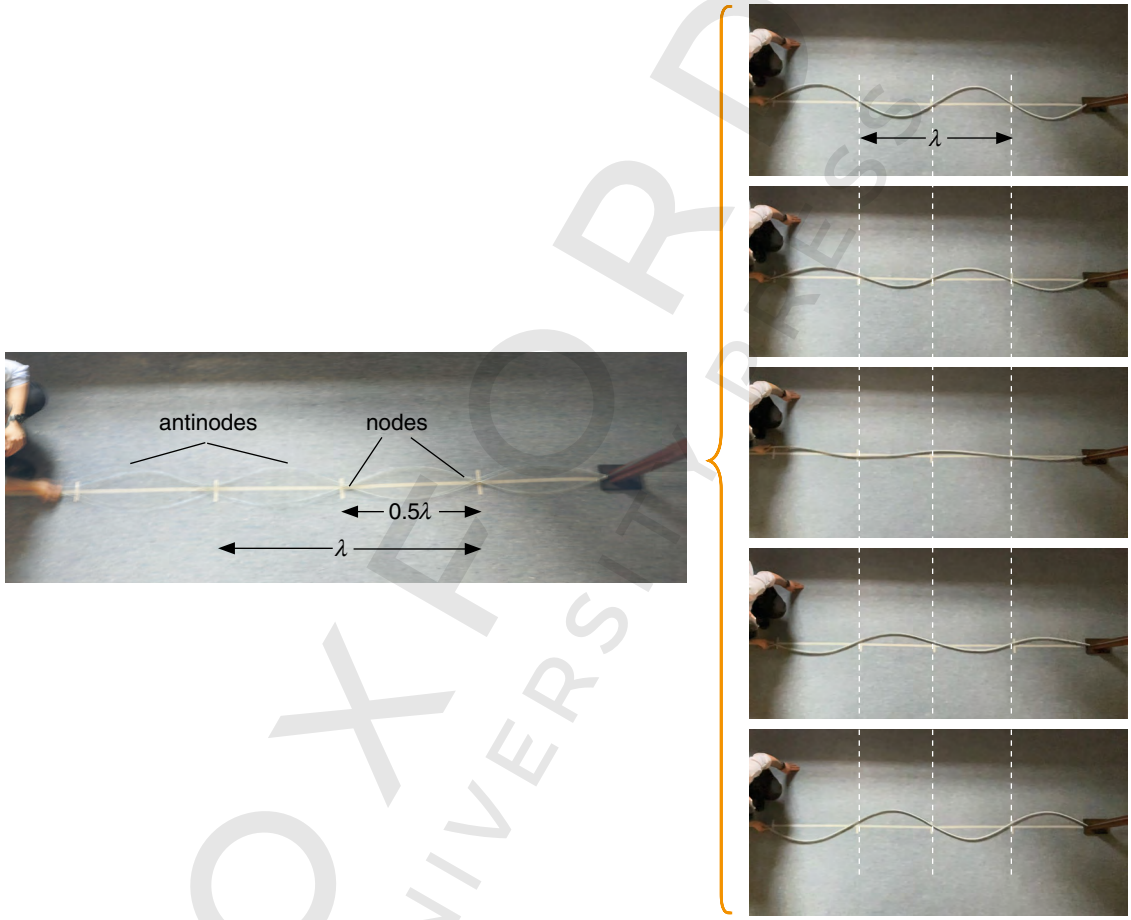


Fig 5.4b *A stationary wave formed on a long spring.*

Some points on a stationary wave do not vibrate. They are called **nodes**. Midway between the nodes are points which vibrate with the largest amplitude. They are called **antinodes**. **The distance between two neighbouring nodes (or antinodes) equals half a wavelength (0.5λ).**

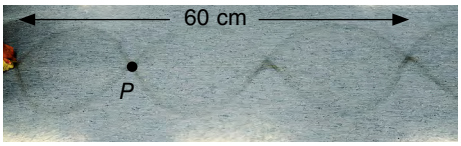
Unlike a travelling wave, energy is not transmitted from one place to another in a stationary wave. Energy is confined within the wave region.

Quick check 13

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 A stationary wave is formed in a long spring.



(a) Find the wavelength of the wave.

(b) Does point *P* on the spring move?

(c) Does the wave carry energy?

(d) Does the wave transmit energy from one place to another?

node 波節 antinode 波腹

DSE goal

Explain the formation of stationary wave.

DSE exam

12(1A)Q16, 15(1A)Q14

2 Formation of a stationary wave

We can see in Experiment 5h that a crest is reflected as a trough at the fixed end (Fig 5.4c). A continuous wave is reflected at a fixed end in the same way (Fig 5.4d). The speed, frequency and wavelength of the wave remain unchanged after the reflection.

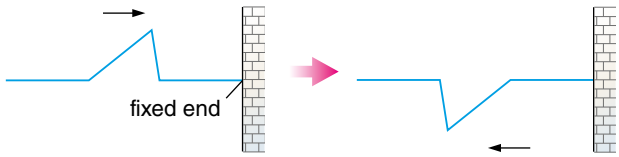


Fig 5.4c *The reflection of a pulse at a fixed end.*

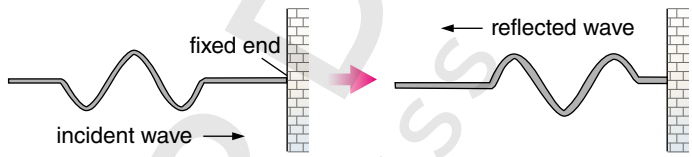


Fig 5.4d *The reflection of a wave at a fixed end.*

As the incident wave and reflected wave cross each other, superposition occurs. The two waves add up and produce a resultant wave. With some particular wavelengths, stationary waves will be formed (Fig 5.4e). With other wavelengths, the string will vibrate irregularly.

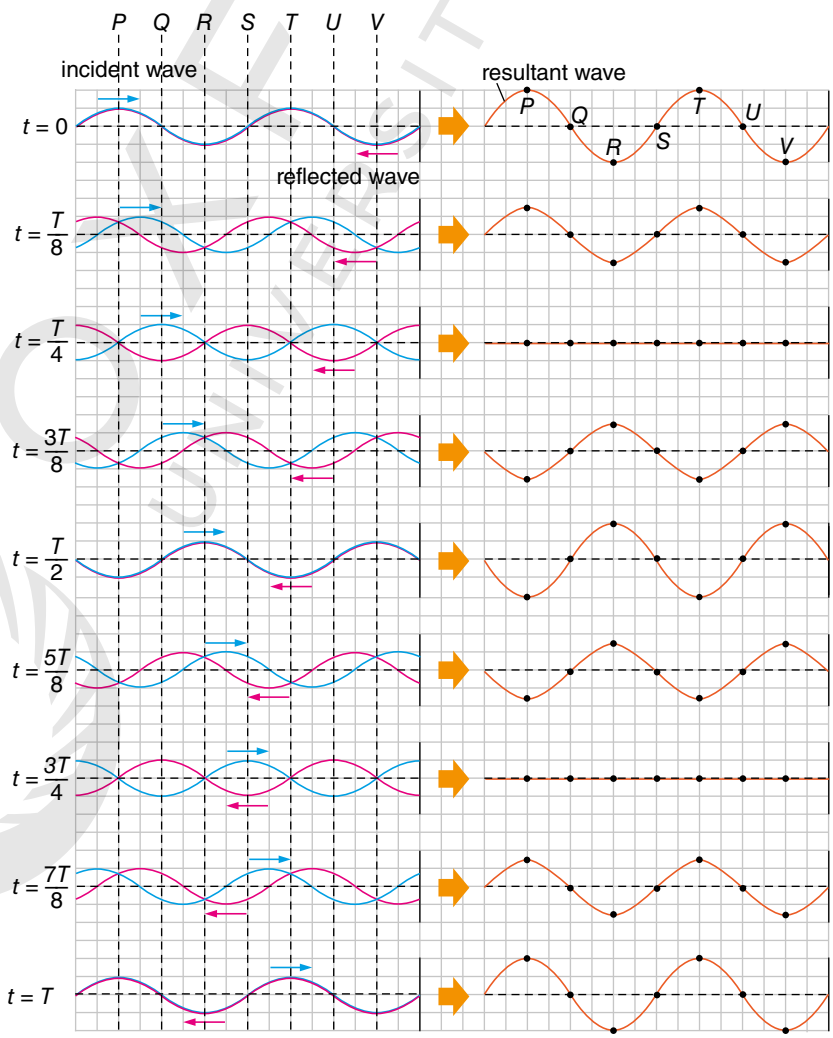
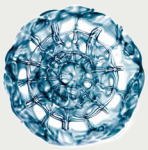


Fig 5.4e *Stationary wave formed by the superposition of incident and reflected waves.*

Know more

Stationary waves on a surface

Stationary waves can also be formed on a surface.



▸ p.99

We can see the following from Figure 5.4e:

1 At points *P*, *R*, *T* and *V*, the incident wave and the reflected wave are always in phase. In the resultant wave, these points vibrate with the largest amplitude. They are antinodes (Fig 5.4f).

2 At points *Q*, *S* and *U*, the incident wave and the reflected wave are always in antiphase. In the resultant wave, these points remain stationary. They are nodes.

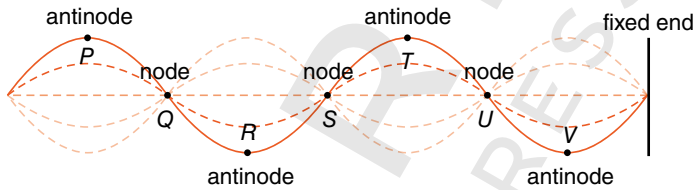


Fig 5.4f *Nodes and antinodes on a stationary wave.*

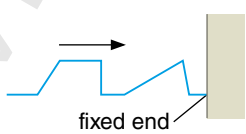
In general, a stationary wave results from the interference of two waves with the same frequency and amplitude travelling at the same speed in opposite directions.

Google Forms Microsoft Forms

Quick check 14

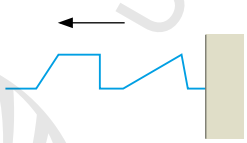
→ E-version in OUP Exercise Platform is also available.

1 Two pulses are travelling on a string with its end fixed.

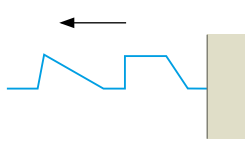


Which of the following best represents the reflected pulses?

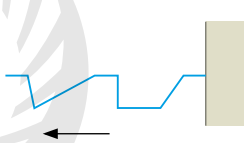
A



B



C



2 A stationary wave is formed between two fixed ends. Which of the following is **not** a possible waveform of the wave?

A



B

C



DSE goal

Describe the characteristics of stationary waves.

DSE exam

15(1A)Q18, 18(1A)Q18, 21(1A)Q15

3 Frequencies of stationary waves

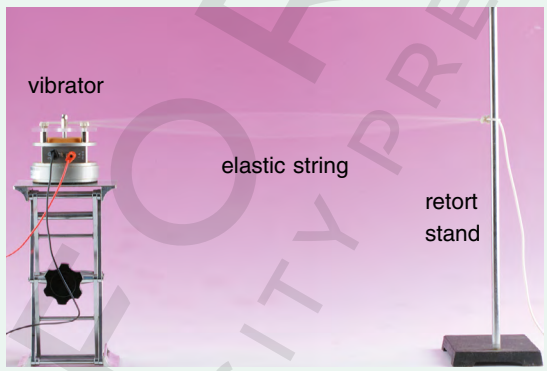
Stationary waves only form at certain frequencies (or wavelengths). We will explore this condition using an elastic string.

→ This is a Video Quiz about this experiment.

Experiment 5i / Frequencies of stationary waves

1 Attach one end of an elastic string to a vibrator and the other end to a retort stand. Make sure that the string is stretched. Connect a signal generator to the vibrator in order to control the vibrating frequency.

The frequency of the vibrator can be read more accurately if we connect the signal generator to a CRO simultaneously. CRO will be introduced to Ss on p.171.



A stroboscope can be used to study the actual appearance of the stationary wave. If the lamp flashes at the same frequency as the wave (or the frequency of the wave is a multiple of that of the lamp), the motion of the wave will be 'frozen'.

2 Slowly increase the frequency of the vibrator until a stationary wave of one loop is produced. Record the frequency.

3 Further increase the frequency until stationary waves of 2, 3 and 4 loops are produced. Record the frequencies.

Discussion

How does the number of loop vary with the frequency of the signal generator?

In Experiment 5i, a stationary wave cannot be produced at most frequencies. Instead, the string just vibrates in an irregular way.

Stationary waves only form at certain frequencies (Fig 5.4g). If a wave of one loop is produced at frequency f, a wave of two loops will be produced at frequency $2f$, and so forth. Example 8 on p.81 verifies this relationship.

Compared to the large vibrations that occur at antinodes, the small vibration at the end connected to the vibrator is negligible. Therefore, the two ends of the string are taken as nodes.

Video

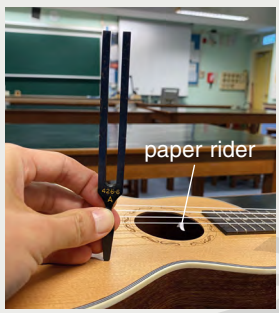
→ This video demonstrates Expt 5i.

→ This video demonstrates how a stroboscope can be used to 'freeze' the motion of a wave.

Know more

Tuning a guitar

You can tune a guitar by using a tuning fork and a paper rider.



▸ p.99

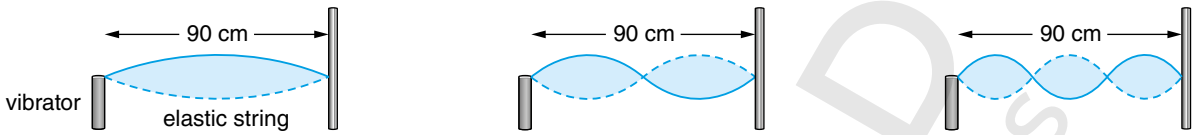


(i) One loop. *(ii) Two loops.* *(iii) Three loops.*

Fig 5.4g *Stationary waves with a different number of loops on an elastic string.*

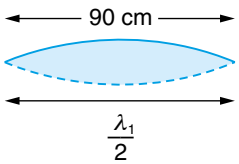
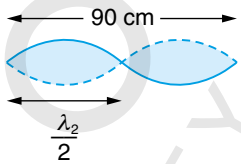
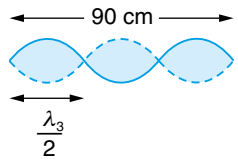
Example 8 | Wavelength, frequency and number of loops

A student produces stationary waves with 1, 2 and 3 loops on an elastic string as shown. The speed of waves along the string is 10.8 m s^{-1}.



Find the wavelength and frequency of each wave. Hence, suggest the relationship between the frequency of the wave f_n and the number of loops n.

Solution

Waveform			
Number of loop (n)	$n = 1$	$n = 2$	$n = 3$
Wavelength (λ_n)	$\frac{\lambda_1}{2} = 90$ $\lambda_1 = 180$ cm	$2 \times \frac{\lambda_2}{2} = 90$ $\lambda_2 = 90$ cm	$3 \times \frac{\lambda_3}{2} = 90$ $\lambda_3 = 60$ cm
Frequency (f_n) By $v = f\lambda$,	$f_1 = \frac{v}{\lambda_1} = \frac{10.8}{1.8} = 6$ Hz	$f_2 = \frac{v}{\lambda_2} = \frac{10.8}{0.9}$ $= 12$ Hz $(= 2f_1)$	$f_3 = \frac{v}{\lambda_3} = \frac{10.8}{0.6}$ $= 18$ Hz $(= 3f_1)$

Therefore, $f_n = nf_1$

When there is only one loop ($n = 1$), the string vibrates in the fundamental mode. The frequency is the lowest and is called the fundamental frequency or first harmonic. Other possible frequencies are called second harmonic (for $n = 2$), third harmonic (for $n = 3$), etc.

Think about...

Can you express λ_n in terms of the length of the string L and the number of loops n?

Google Forms Microsoft Forms

Quick check 15

→ E-version in OUP Exercise Platform is also available.

1 *True or false:* Stationary waves can be formed at any frequency. (T/F)

2 A stationary wave of n loops is formed on a string with both ends fixed. Its wavelength is λ, frequency is f and speed is v. The length of the string is L. Complete the table below.

	n	λ / cm	f / Hz	v / m s^{-1}	L / cm
(a)	1	10	5		
(b)			16	8	150

Learn by practice 5

(For Q1–2.) A student connects an elastic string to a vibrator. The other end is connected to a fixed support. A stationary wave is formed on the string when the vibrator vibrates at frequency f.

1 When f = 20 Hz, the wave has 5 loops. Find the number of loops when f = 40 Hz.

2 The string is 75 cm long and the speed of waves along the string is 12 m s^{-1}. Find the wavelengths and frequencies when stationary waves of (a) 2 loops, (b) 3 loops and (c) 5 loops are produced.

Writing practice

One end of a string is tied to a vibrator and the other end is fixed. A stationary wave is formed on the string when the vibrator vibrates.

Question 1

Explain why a stationary wave is formed on the string. (4 marks)

Answer

The incident wave produced by the vibrator is reflected at the fixed end. 1A

The reflected wave and the incident wave travel at the same speed in opposite directions, 1A

and have the same frequency and amplitude. 1A

They add up and a stationary wave is formed if the frequency is a certain value. 1A

Question 2

The tension in the string is gradually reduced until a new stationary wave is formed. Explain whether the new wave has more or fewer loops. (3 marks)

Answer

When the tension decreases, the wave speed v decreases. 1A

By $v = f\lambda$, for constant f, λ decreases when v decreases. 1A

As a result, the new stationary wave has more loops. 1A

Practise yourself

1 State the conditions for a stationary wave to be formed on a string. (3 marks)

2 One end of a string is tied to a vibrator and the other end is fixed. A stationary wave is formed on the string. The tension in the string is varied such that a new stationary wave with fewer loops is formed. Explain whether the tension increases or decreases. (3 marks)

DSE goal

Describe the characteristics of stationary waves.

DSE exam

12(1A)Q18, 13(1A)Q18, 16(1A)Q18

4 Particle vibrations of stationary waves

Let us take a closer look at the particle vibrations of a stationary wave.

1 **Particles, except those at nodes,** vibrate with the **same frequency**, i.e. they complete one cycle of vibration in the same period of time.

2 Particles at **different positions** in the same loop vibrate with **different amplitudes** (Fig 5.4h).

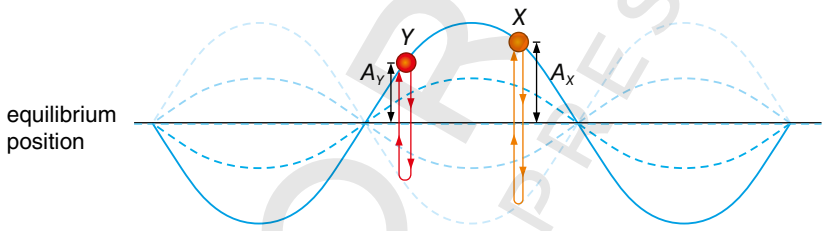


Fig 5.4h *Particles X and Y on a stationary wave vibrate with different amplitudes.*

3 Particles in the **same loop** (e.g. *M*, *N* and *O*) vibrate **in phase**, while particles in two **adjacent loops** (e.g. *O* and *Q*) vibrate **in antiphase** (Fig 5.4i and 5.4j). All particles reach their respective maximum displacements at the same time.

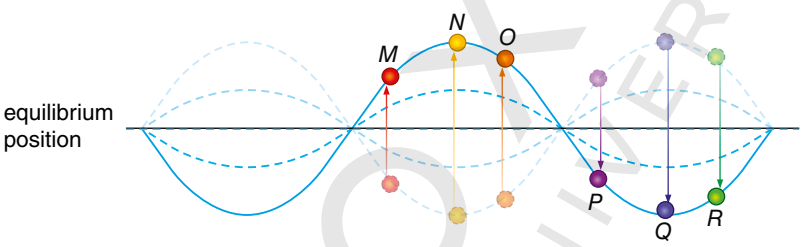


Fig 5.4i *Particles in the same loop vibrate in phase while particles in two adjacent loops vibrate in antiphase.*

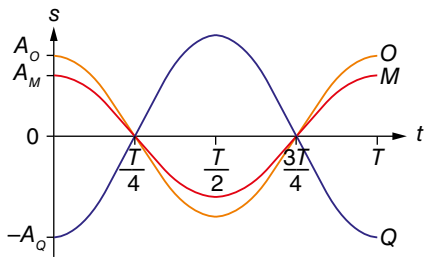


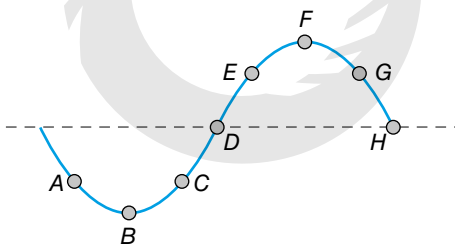
Fig 5.4j *Displacement–time graphs of particles M, O and Q.*

Quick check 16

→ E-version in OUP Exercise Platform is also available.

Google Forms | Microsoft Forms

1 A stationary wave on a string at a certain instant is as shown. *A* to *G* are particles on the string.



At this instant, *E* is moving upwards. Determine whether the following statements are true.

(a) *B* vibrates with the same frequency as *E*. (T/F)

(b) *B* vibrates with same amplitude as *E*. (T/F)

(c) *A*, *B* and *C* vibrate in phase. (T/F)

(d) *B* vibrates in antiphase with *E*. (T/F)

(e) *F* is momentarily at rest at the instant shown. (T/F)

(f) *D* remains at rest. (T/F)

(g) *C* and *F* return to their equilibrium positions at the same time. (T/F)

We can see that stationary waves are different from travelling waves. Table 5.4a shows a comparison between the two types of waves.

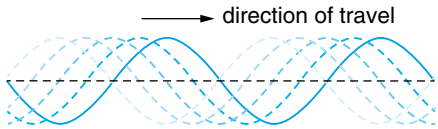
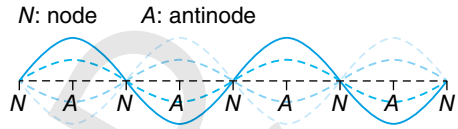
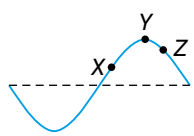
	Travelling wave	Stationary wave
		
Propagation	moves forwards	does not move forwards
Wavelength	the minimum distance the waveform repeats itself	the total length of two loops
Amplitude	same for all particles	different for different particles
Frequency	same for all particles	
Particle motion	different particles reach their maximum displacements at different times	all particles reach their respective maximum displacements at the same time
Phase	• particles $n\lambda$ apart are in phase • particles $(n - 0.5)\lambda$ apart are in antiphase • neighbouring particles are out of phase	• particles in the same loop are in phase. • particles in adjacent loops are in antiphase
Energy	transmitted from one place to another	not transmitted from one place to another

Table 5.4a *Comparison between travelling and stationary waves.*

Remind Ss that a stationary wave have the same frequency, wavelength and wave speed as the travelling waves that form it.

Example 9 Particle motion on a wave

The figure shows a wave on a string at time $t = 0$. X, Y and Z are particles on the string. At this instant, X and Z are both moving downwards.

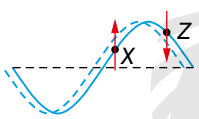


(a) Is this a travelling wave or stationary wave?

(b) The period of the wave is T. On the same figure, sketch the displacement–time graphs for X and Y from $t = 0$ to $t = T$.

Tip

If this were a travelling wave, X and Z would move in opposite directions.



Solution

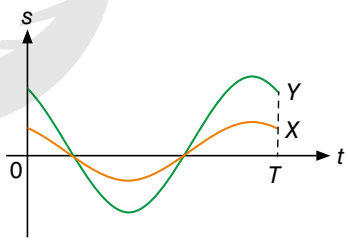
(a) Stationary wave

Tip

Since X is moving downwards at $t = 0$, it is not at the maximum displacement. s_X is smaller than the amplitude and is decreasing.

As Y vibrates in phase with X, its s–t graph has the same shape but with a larger amplitude.

(b)



Revision 5 Q29 (p.96)

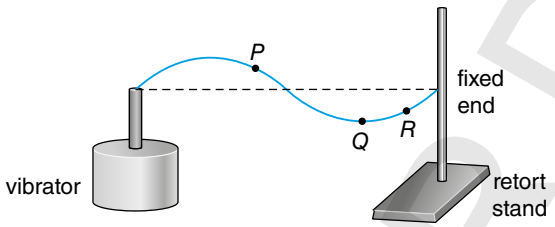
☆ HKDSE 2016 Paper 1A Q18

→ This video demonstrates how this question is solved.

Video

DSE practice 3 | Particles on a stationary wave

One end of a string is tied to a vibrator and the other end is fixed to a retort stand. When the vibrator vibrates at a certain frequency, a stationary wave is formed. The figure below shows the waveform at a certain instant. *P*, *Q* and *R* are three points on the string.



Which of the following statements must be correct?

(1) If the frequency of the vibrator halves, the wave speed will halve.

(2) *Q* is momentarily at rest at this instant.

(3) If the frequency of the vibrator doubles, *P* and *R* will vibrate in phase.

A (2) only

B (3) only

C (1) and (2) only

D (1) and (3) only

Solution

The wave speed is independent of the frequency.

Common mistake ⚠
Students may wrongly think that the wave speed will halve.

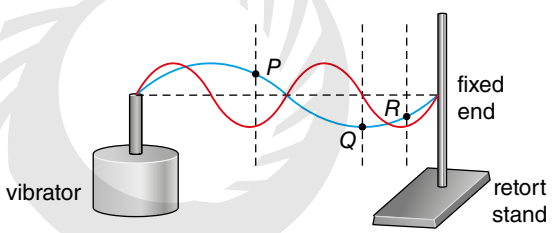
∴ (1) is incorrect.

We cannot tell whether the particles reach their maximum displacements or not.

∴ (2) is not necessarily correct.

When the frequency doubles, the wavelength will halve.

∴ There will be 4 loops.



∴ *P* and *R* will vibrate in phase.

Q will be at a node and remain stationary.

∴ (3) is correct.

∴ The answer is B.

▶ Revision 5 Q17 (p.93)

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Exercise 5.4

★**1** Two pulses on a string move towards each other as shown (Fig a).

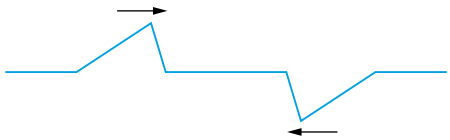


Fig a

When the pulses meet, which of the following statements about the displacement of a point on the resultant pulse is/are correct?

(1) It is the sum of the displacements due to both pulses at that point.

(2) It is the sum of the amplitudes of both pulses.

(3) It can never be zero.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

★**2** Figure b shows a stationary wave produced by a vibrator with frequency 18 Hz. By changing the frequency of the vibrator, stationary waves with different numbers of loops can be produced.

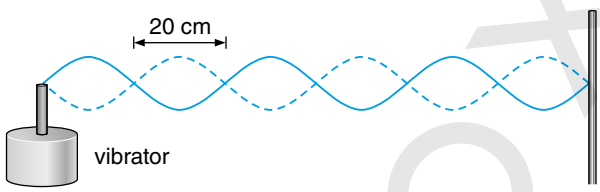


Fig b

Which of the following is/are the possible wavelength(s) of the stationary waves that can be produced?

(1) 10 cm

(2) 60 cm

(3) 240 cm

A (1) only B (2) only

C (1) and (2) only D (1), (2) and (3)

★**3** A vibrator with frequency f produces a stationary wave with 3 loops on a string (Fig c).

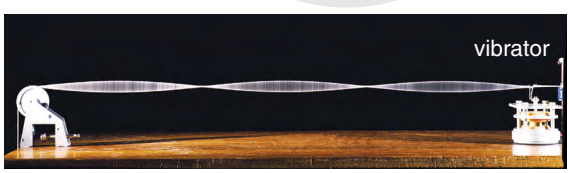


Fig c

What happens to the wave when the frequency of the vibrator becomes $1.5f$?

A The speed of the wave becomes 1.5 times of the original value.

B A stationary wave with 2 loops is produced.

C A stationary wave with 6 loops is produced.

D No stationary wave is produced.

★**4** Figure d shows a stationary wave produced by a vibrator with frequency f. P to S are four particles on the wave.

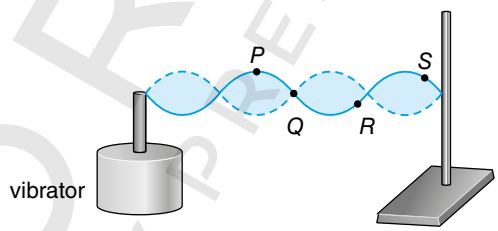


Fig d

(a) Compare the frequencies of the vibrations of P and R.

(b) Compare the amplitudes of the vibrations of P and R.

(c) P is moving upwards at a certain instant. How do Q, R and S move at this instant?

(d) Determine whether P and R vibrate in phase or in antiphase if the frequency of the vibrator becomes (i) $0.5f$, (ii) $0.25f$.

★**5** A stationary wave is formed on a string by two identical waves travelling in opposite directions. At the instant shown in Figure e, particle X on the string attains its maximum displacement. It vibrates at 20 Hz.

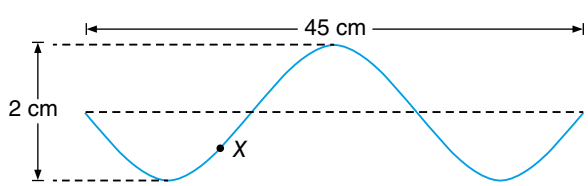


Fig e

(a) Find the amplitude, wavelength and speed of each travelling wave.

(b) Compare a travelling wave and a stationary wave with regard to

(i) the transmission of energy,

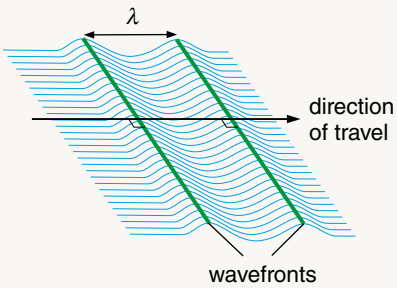
(ii) the transfer of matter.

Key concepts 5

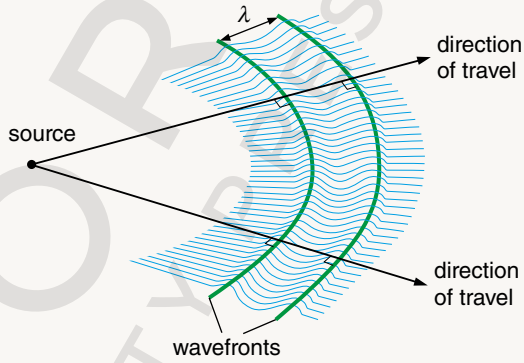
1 Wavefronts

- Particles on the same wavefront vibrate in phase.
- Wavefronts are always perpendicular to the direction the wave travels.

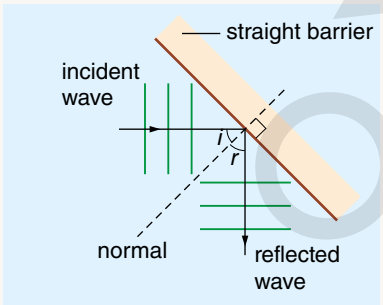
Straight wavefronts

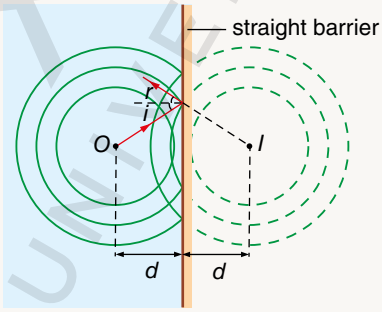


Circular wavefronts



2 Reflection



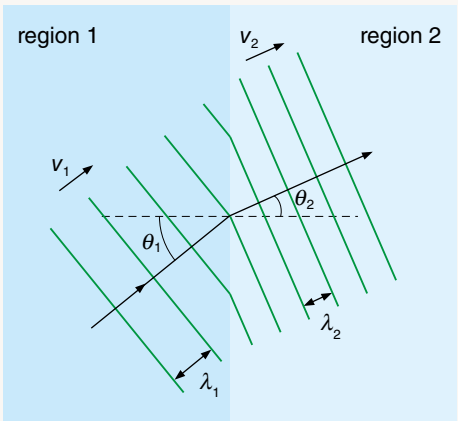


- $r = i$
- v, f, λ remain unchanged

3 Refraction

- Refraction is the change in a wave's travelling direction due to a change in wave speed v.

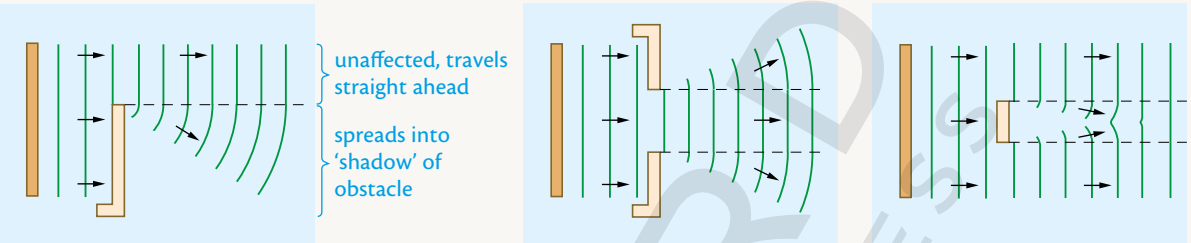
  When $v \downarrow, \lambda \downarrow$ but f remains unchanged
- For water waves, water depth $\downarrow \Rightarrow v \downarrow$
- $\dfrac{\sin \theta_1}{\sin \theta_2} = \dfrac{v_1}{v_2} = \dfrac{\lambda_1}{\lambda_2}$



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4 Diffraction

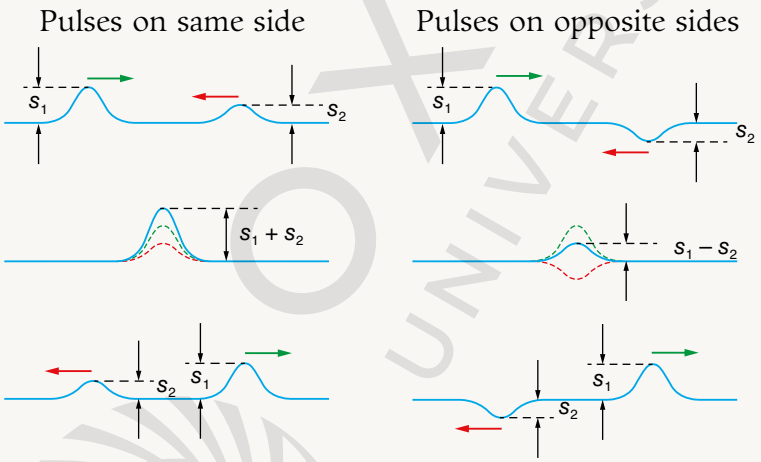
- Diffraction is the bending or spreading of a wave around the edge into the 'shadow' of an obstacle.



- v, f, λ remain unchanged
- Degree of diffraction ↑ when
  - gap width ↓
  - λ ↑

5 Interference

- When two pulses/waves meet, their displacements add up.

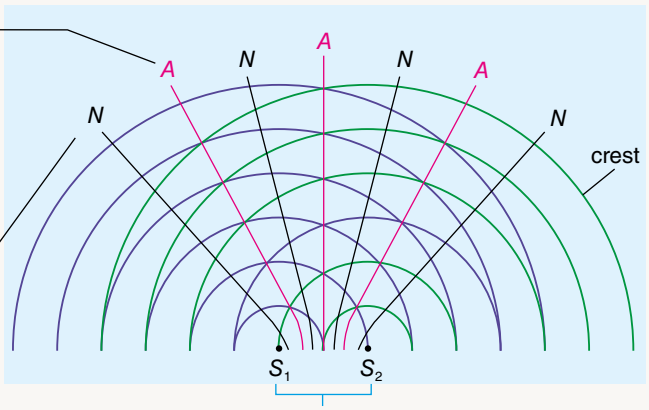


antinodal lines

- constructive interference
- particles vibrate with max. amplitude
- Path difference $\Delta = n\lambda$

nodal lines

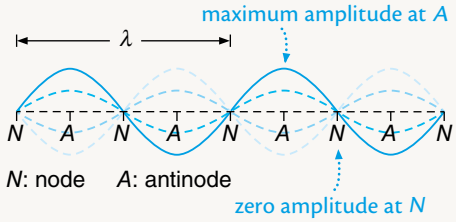
- destructive interference
- particles remain at rest
- $\Delta = (n + 0.5)\lambda$

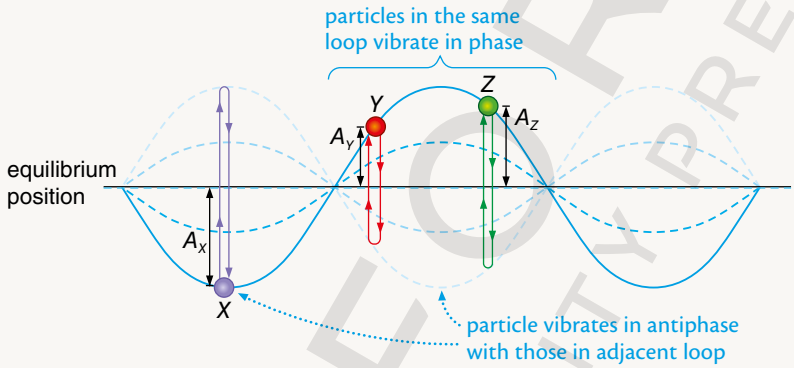


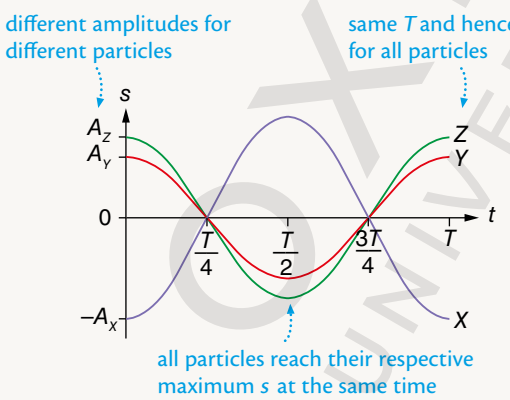
coherent wave sources can produce a steady interference pattern

6 Stationary waves

- Result from the interference of two waves with the same f, λ and v travelling in opposite directions.







Key terms

1	antinodal line 腹線	p.68
2	antinode 波腹	p.77
3	circular wave 圓形波	p.40
4	coherent sources 相干源	p.63
5	constructive interference 相長干涉	p.64
6	destructive interference 相消干涉	p.65
7	diffraction 衍射 / 繞射	p.59
8	interference 干涉	p.63
9	nodal line 節線	p.68
10	node 波節	p.77
11	path difference 程差	p.66
12	plane wave 平面波	p.40
13	ripple tank 水波槽	p.39
14	stationary wave / standing wave 駐波	p.76
15	straight wave 直線波	p.40
16	superposition 疊加	p.62
17	wavefront 波陣面	p.40

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Revision 5

Multiple-choice questions

1 Misc A straight water wave travels towards a straight barrier (Fig a).

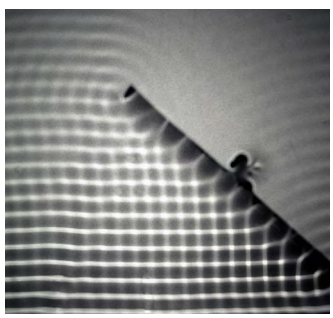


Fig a

Which of the following phenomena of a wave can be observed in the figure?

(1) Reflection
(2) Diffraction
(3) Interference

A (1) only
B (2) only
C (1) and (2) only
D (1), (2) and (3)

***2** 5.2 The figure shows the water wave reflected by a barrier (Fig b).

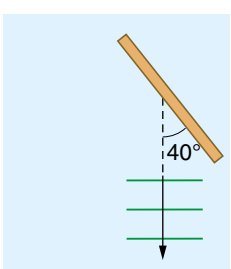
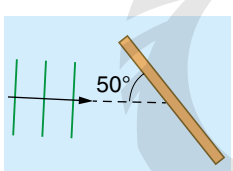
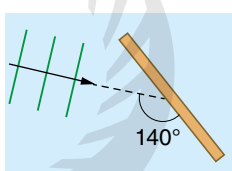
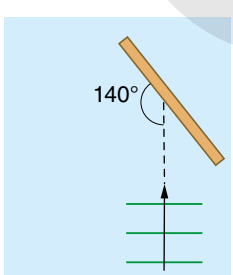


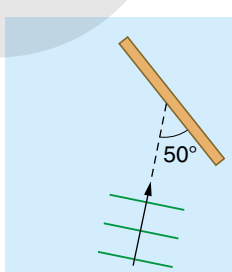
Fig b

Which of the following diagrams correctly shows the incident wave?

A


B


C


D


***3** 5.3 A water wave with a wavelength of 3 cm and a frequency of 5 Hz travels through a gap (Fig c). How long does the wave take to travel from point *X* to point *Y*?

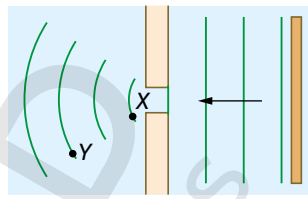


Fig c

A 0.2 s
B 0.4 s
C 0.6 s
D Cannot be determined

***4** 5.3 Two pulses travel towards each other on a string as shown (Fig d). The pulses overlap with each other completely at *X* at time $t = t_0$.

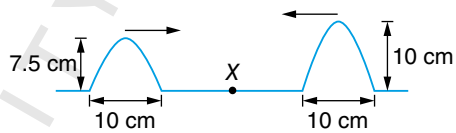
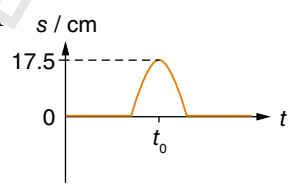
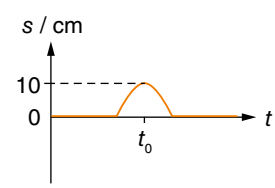
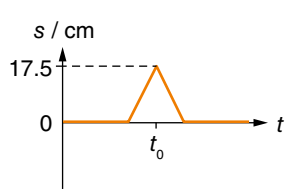


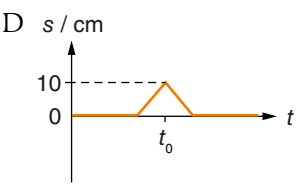
Fig d

Which of the following graphs best shows how the displacement *s* at point *X* changes with *t*? Take the upward direction as positive.

A


B


C


D


***5** 5.2 A straight wave travels from medium *X* into medium *Y* at an angle. Its wavelength shortens in medium *Y*. Which of the following statements is/are correct?

(1) The wave speed is higher in *Y* than in *X*.
(2) The wave bends towards the normal when crossing the boundary.
(3) The wave will bend more if the change in wave speed is larger.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

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★6 Misc Figure e (not drawn to scale) shows how a water wave enters and leaves region *X*.

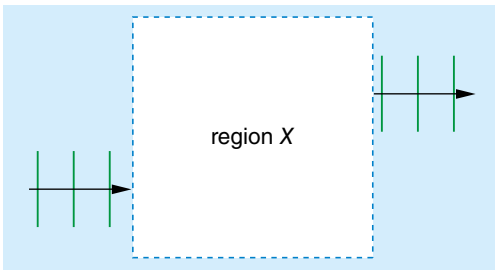
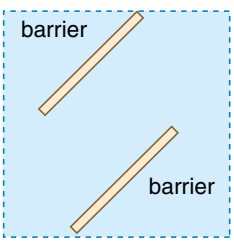


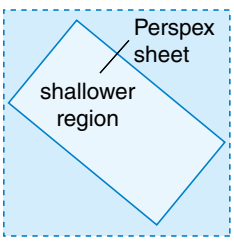
Fig e

Which of the following can be the structure in region *X*?

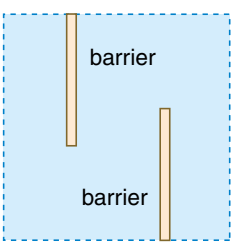
(1)



(2)



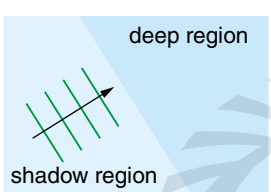
(3)



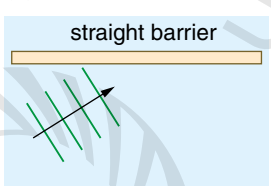
A (1) only
B (1) and (2) only
C (2) and (3) only
D (1), (2) and (3)

★7 5.2 Which of the following situations will cause the water wave to change its travelling direction?

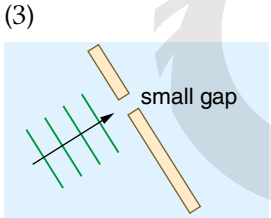
(1)



(2)



(3)



A (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

★8 5.1

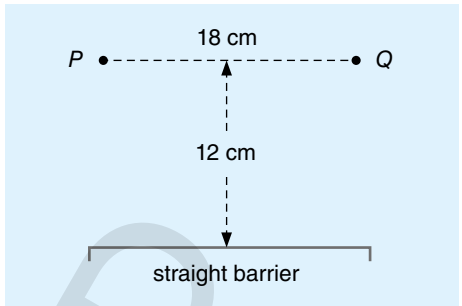


Fig f

Figure f shows a ripple tank with a straight barrier. Positions *P* and *Q* are 18 cm apart and are both 12 cm from the straight barrier. A cork is placed at position *Q*. When a point dipper at *P* touches the water surface once, a pulse is produced. The cork at *Q* is set into vibration twice. Given that the speed of the water wave in the ripple tank is 0.3 m s^{-1}, find the time interval between the two vibrations of the cork. Assume that the pulse is only reflected by the barrier.

A 0.2 s
B 0.4 s
C 1 s
D 1.4 s

★9 5.3 Two dippers, S_1 and S_2, vibrate in phase and produce water waves in a ripple tank (Fig g). *MN* is the perpendicular bisector of S_1S_2.

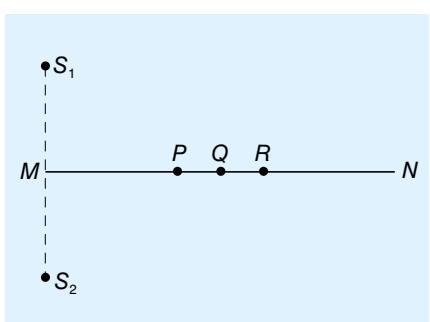


Fig g

Which of the following must be correct?

(1) Constructive interference occurs at *P*, *Q* and *R*.

(2) The particles at *P*, *Q* and *R* reach their maximum displacements at the same time.

(3) The distance between *Q* and S_1 is equal to a whole number of wavelengths.

A (1) only
B (2) only
C (1) and (3) only
D (1), (2) and (3)

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★10 5.4 Figure h shows a stationary wave produced on a string by a vibrator with a frequency of 24 Hz.

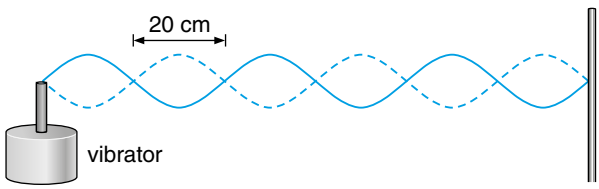


Fig h

Which of the following frequencies of the vibrator **cannot** produce a stationary wave on the string?

A 2 Hz B 4 Hz
C 8 Hz D 16 Hz

★11 Misc

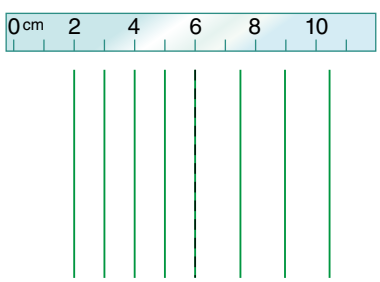


Fig i

A straight water wave travels from a shallow region to a deep region in a ripple tank (Fig i). If the wave speed in the deep region is 12 cm s^{-1}, what is the wave speed in the shallow region?

A 2 cm s^{-1} B 8 cm s^{-1}
C 12 cm s^{-1} D 18 cm s^{-1}

★12 5.4 A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave of n loops is formed when the vibrator vibrates with frequency f. Which of the following statements is/are correct?

(1) If the frequency of the vibrator is increased to $2f$, the speed of the wave in the string will be doubled.

(2) If the frequency of the vibrator is increased to $2f$, the stationary wave will have $2n$ loops.

(3) If the tension in the string is increased gradually, the stationary wave formed will have more loops.

A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

★13 5.4 Figure j shows the s–t graph of a particle P on a stationary wave formed on a string. The period of the wave is t_0.

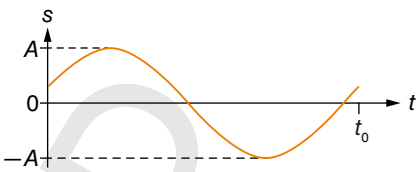
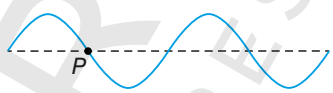


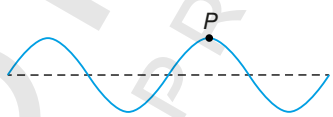
Fig j

Which of the following shows the possible position of P on the wave at $t = 0$?

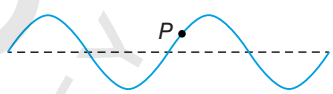
(1)



(2)



(3)



A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

★★14 5.3 Two coherent sources, X and Y, generate circular waves of wavelength 3 cm in a ripple tank (Fig k). The distance between X and Y is 17 cm. How many points on the dotted straight line between X and Y show constructive interference?

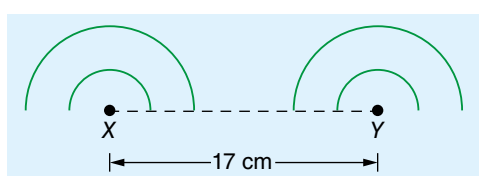


Fig k

A 5 B 6
C 11 D 12

Refer p.73

15 5.4 **HKDSE 2015 Paper 1A Q14**

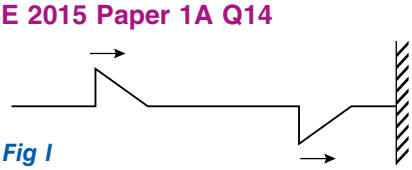
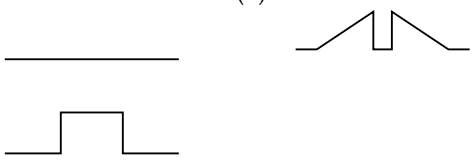


Fig l

Two pulses of the same shape travel along a stretched string with one end fixed to the wall as shown above (Fig l). Which of the following can be the resultant waveform at different instants later?

Q15 Exam report: Almost 60% of candidates had difficulty with this question which suggests that a majority were not familiar with the reflection of wave pulses at a fixed end.

(1) (2)



(3)

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only (41%)

16 HKDSE 2015 Paper 1A Q18

5.4

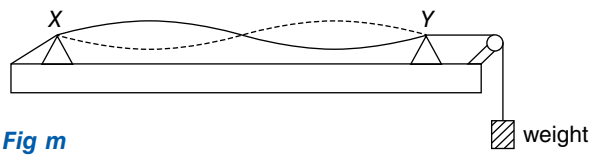
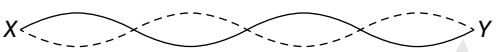
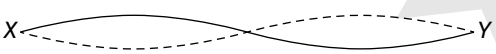


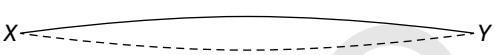
Fig m

A string is set to vibrate at frequency f such that a standing wave is formed between two fixed supports X and Y as shown (Fig m). If the tension in the string is increased by adding weight gradually while the frequency is kept at f, which of the following is a possible mode of vibration at a steady state?

A 

B 

C 

D 

(61%)

17 HKDSE 2016 Paper 1A Q18

5.4

A string is tied to a vibrator while the other end is fixed to a wall. A stationary wave is formed as shown (Fig n).

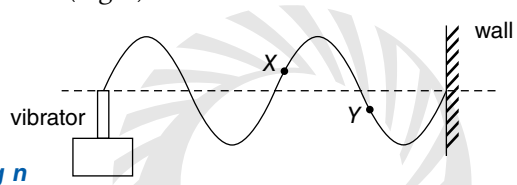


Fig n

Which statement is correct when the frequency of the vibrator doubles?

A The wavelength will double.

B The wave speed will double.

C The amplitude will be halved.

D Particles X and Y will become vibrating in phase. (49%)

Q17 Exam report: About one-third of the candidates thought that the wave speed will double as the frequency doubles (option B).

18 HKDSE 2019 Paper IA Q16

5.3

Figure o shows the circular water waves generated by two dippers S_1 and S_2 vibrating in phase. The lines represent wave crests.

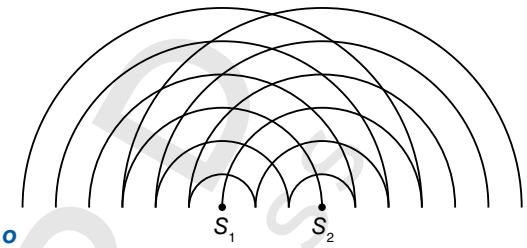


Fig o

What is the number of nodal lines (i.e. minimum amplitude) formed?

A 3

B 4

C 6 (56%)

D 7

19 HKDSE 2021 Paper 1A Q14

5.3

In a ripple tank, circular water waves of wavelength λ_1 and λ_2 ($\lambda_1 > \lambda_2$) are produced by two vibrators S_1 and S_2 respectively. Figure p represents the two circular waves propagating on the water surface at a certain moment.

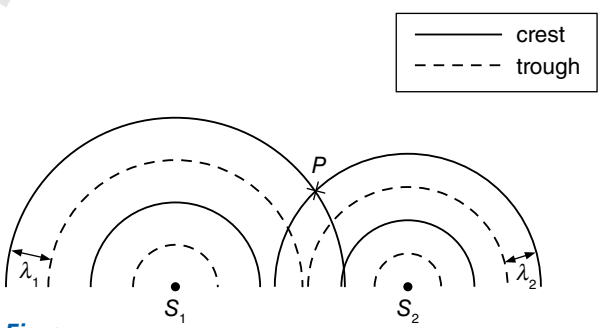


Fig p

Which of the following statements is correct?

A The particle at P is always at crest position.

B At P, the two waves always reinforce to give a larger amplitude.

C The principle of superposition cannot be applied at P as $\lambda_1 \neq \lambda_2$.

D The principle of superposition can be applied at P but the two waves do not always reinforce at that location. (47%)

Q19 Exam report: About 30% of the candidates wrongly held that two waves of different frequencies would always reinforce to give a larger amplitude at a certain location.

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Conventional questions

★**20** Misc A boat is positioned in the sea as shown (Fig q). A transverse wave is travelling towards the shore.

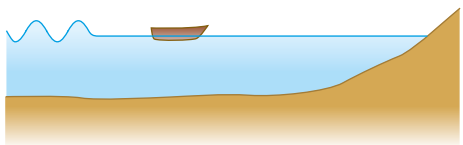


Fig q

(a) Describe the motion of the boat when a water wave travels past it. (1 mark)

(b) How do the wave speed, wavelength and frequency of the wave change when the wave approaches the shore? (3 marks)

(c) A tsunami is an extremely large wave that occurs when a large body of water is disturbed vertically. It can be caused by an earthquake, a volcanic eruption or even a landslide.

(i) A tsunami with wavelength of 100 km and a period of 8 minutes travels across the ocean. Find the speed of this tsunami. (2 marks)

(ii) Sometimes when a tsunami travels across the ocean, its wavefronts bend. Suggest a possible reason. (1 mark)

★**21** 5.1 You are given a rectangular tray filled with water, a ruler, and a stop-watch as shown in Figure r.

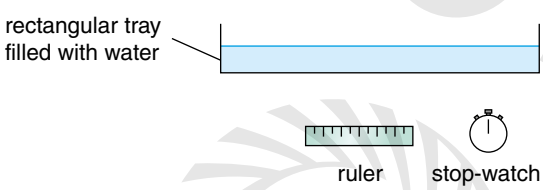


Fig r

(a) Describe how you would use the above apparatus to find the speed of a pulse travelling in the tray of water. (4 marks)

(b) Suggest one possible error in the setting. (1 mark)

(c) Which tray, the one with deeper water or the one with shallower water, will have a smaller percentage error when finding the speed of the pulse travelling in it? Explain your answer. (3 marks)

★**22** Misc A series of water waves with a frequency of 5 Hz travels in a water tank (Fig s). The tank is divided into two regions, *A* and *B*, of different depths. The wavelength in region *A* is 0.2 m and that in region *B* is 0.24 m.

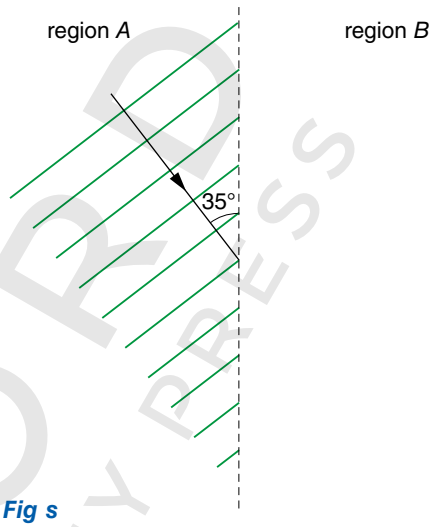


Fig s

(a) Find the wave speed of the water waves in region *A*. (2 marks)

(b) Find the angle of refraction of the wave in region *B*. (2 marks)

(c) Which region is deeper? (1 mark)

(d) Sketch the refracted wavefronts in region *B*. (2 marks)

(e) If a wave travels from region *B* to region *A* with an angle of incidence the same as the value found in (b), what will be the angle of refraction? (1 mark)

★**23** Misc In a ripple tank, a barrier is placed at the boundary of the shallow region and the deep region (Fig t). The vibrator then vibrates to produce straight waves. The wave travels at 0.3 m s^{-1} in the shallow region and 0.4 m s^{-1} in the deep region. The wavelength of the wave in the shallow region is 1.5 cm.

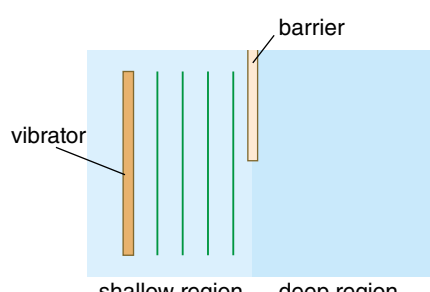


Fig t

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(a) Sketch the wave pattern in the deep region. (2 marks)

(b) Explain the wave pattern drawn in (a). (2 marks)

(c) Find the frequency of the waves in the deep region. (2 marks)

(d) Find the wavelength of the waves in the deep region. (2 marks)

★**24** Misc Tom stands in a swimming pool as shown in Figure u. A toy boat is floating on the water surface. Tom moves his left hand up and down on the water surface to produce a continuous water wave. Assume the wave does not reflect at the side of the pool.

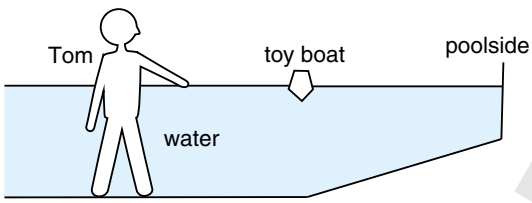


Fig u

(a) Tom thinks that the water wave can push the boat forwards. Explain whether he is correct. (2 marks)

(b) Tom moves both hands up and down in phase to produce water waves. The wavelength is 8 cm before reaching the toy boat. The toy boat is 30 cm away from his left hand and 42 cm from his right hand. Describe and explain the motion of the toy boat. (3 marks)

(c) Then Tom removes the toy boat and produces a continuous straight water wave by using a long foam tube (Fig v). The wave travels towards the side of the pool. Complete the wave pattern in Figure v. (1 mark)

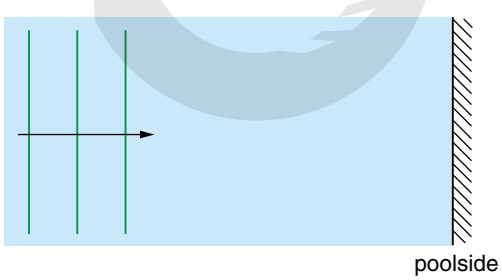


Fig v

★**25** 5.4 Figure w shows the displacement–distance graph of a stationary wave at time $t = 0$. The frequency of the wave is 0.5 Hz. P and Q are particles on the wave. At $t = 0$, P is momentarily at rest.

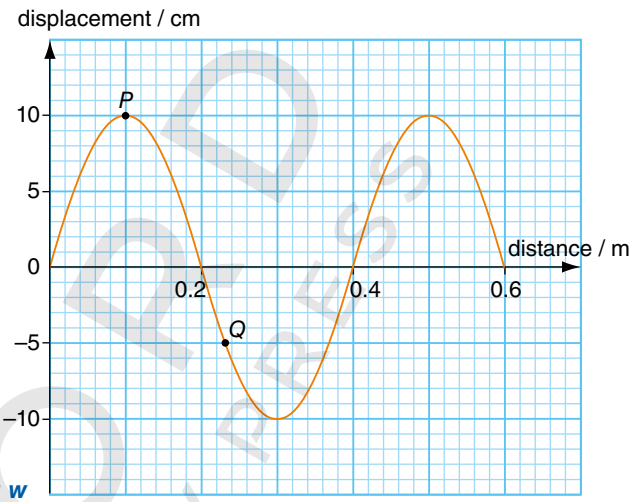


Fig w

(a) Sketch the displacement–time graph of P from $t = 0$ to $t = 3$ s. (3 marks)

(b) Sketch the displacement–time graph of Q from $t = 0$ to $t = 3$ s on the same figure in (a). (2 marks)

(c) The frequency of the wave now becomes 1 Hz and its amplitude remains unchanged. In Figure w, sketch the new displacement–distance graph at an instant that Q is momentarily at rest. (1 mark)

★**26** 5.3 Figure x shows the interference pattern observed in a ripple tank at time $t = 0$. X and Y are coherent point sources vibrating in phase. The solid lines represent crests and the dotted lines represent troughs. The period of the waves is 0.2 s.

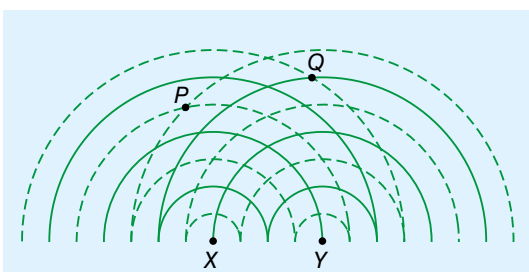


Fig x

(a) Which type of interference occurs at P and Q respectively? (2 marks)

(b) Sketch the s–t graphs of the particles at P and Q respectively from $t = 0$ to $t = 0.4$ s. (4 marks)

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(c) R is a point on the antinodal line representing a path difference of one wavelength. A particle at R reaches the crest at $t = 0$. In Figure x, draw the mentioned antinodal line (label it as L_1) and mark all possible positions for R. (2 marks)

(d) Which type of interference occurs at P and Q respectively if the vibrating frequency of X and Y doubles? (2 marks)

(e) Besides changing the vibrating frequency of X and Y, suggest two other methods that can increase the number of antinodal lines. (2 marks)

*27 5.4 Read the following passage about **wave motion in a tennis racquet** and answer the questions that follow.

When you hit a ball with a tennis racquet, you can feel the vibration of the racquet caused by the impact of the ball. The vibration of the racquet is in fact a stationary wave (Fig y), with one node of the wave near the centre of the network of strings, and another node on the handle. The vibration felt by your hand can be largely reduced if you hold the handle at the right position and strike the ball at the node position on the network of cords.

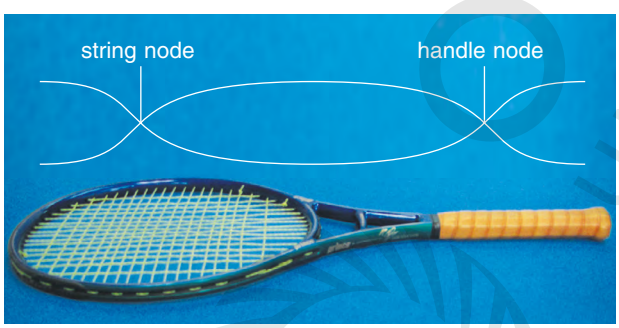


Fig y

(a) If the distance between the two nodes is 0.5 m, what is the wavelength of the stationary wave on the racquet? (1 mark)

(b) Suppose the wave speed along the racquet is 140 m s^{-1}. Find the frequency of the stationary wave. (2 marks)

(c) At which position should a player hold a tennis racquet in order to reduce the feeling of the vibration? (1 mark)

**28 5.2 A straight wave is produced in a ripple tank. The tank has two regions, X and Y, which have different water depths. The wave pattern observed at an instant is shown below (Fig z).

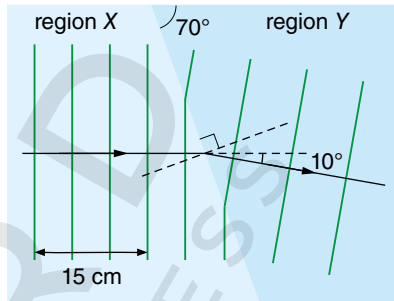


Fig z

(a) What happens to the wave when it comes across the boundary of the two regions? Why does this happen? (2 marks)

(b) Find the wavelength of the wave in X. (1 mark)

(c) Find the wavelength of the wave in Y. (4 marks)

(d) What happens to the wave if the water depth in X is reduced? In Figure z, draw an arrow to show the direction of propagation of the wave in Y. (3 marks)

Refer p.55

**29 5.4

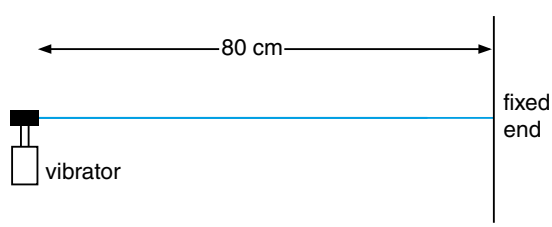


Fig aa

One end of a string is connected to a vibrator and the other end is fixed (Fig aa). The length of the stretched string is 80 cm. The vibrator vibrates to produce a stationary wave on the string.

(a) (i) What is the largest possible wavelength of the stationary wave produced? (1 mark)

(ii) To produce a stationary wave of the largest possible wavelength, the frequency of the vibrator is set to 4 Hz. Find the wave speed along the string. (2 marks)

(b) The frequency of the vibrator is then set to 16 Hz. Figure ab shows the waveform at time $t = 0$. A, B, C, D and E are particles on the stationary wave. At $t = 0$, C is 3 cm below the highest position it can reach, and it is moving downwards.

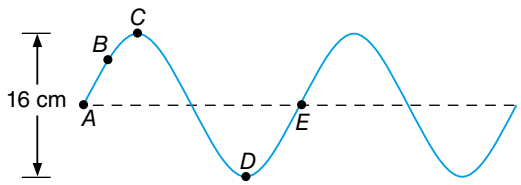


Fig ab

(i) Which particle(s) is/are also moving downwards at $t = 0$? (1 mark)

(ii) Sketch the displacement–time graphs of B and D from $t = 0$ to $t = 2T$. (3 marks)

Refer Eg 9 (p.84)

See p.99 for the Exam Report.

30 HKDSE 2012 Paper 1B Q6 ☆ Part (d) omitted

5.3

In a ripple tank, circular water waves are produced by two vibrators S_1 and S_2 of the same frequency vibrating in phase. Their separation is 3.5λ, where λ is the wavelength of the waves. Figure ac shows the two circular waves propagating on the water surface at a certain moment. Line L is a line connecting all points P which have path difference $S_1P - S_2P = 0$.

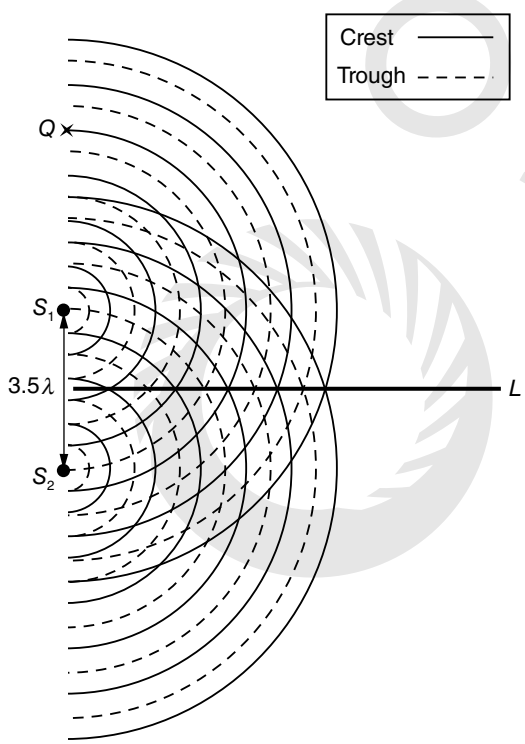


Fig ac

(a) Draw and label a line in Figure ac connecting all points P which have path difference

(i) $S_1P - S_2P = \lambda$ (label it as L_1)

(ii) $S_1P - S_2P = -\frac{3}{2}\lambda$ (label it as L_2)

What would happen to L_1 and L_2 if the separation between S_1 and S_2 is reduced slightly? (3 marks)

(b) Figure ad shows the profile of the water level along line L at a certain instant. Sketch on the same figure the profile at a time $\frac{T}{2}$ later, where T is the period of the water waves. (1 mark)

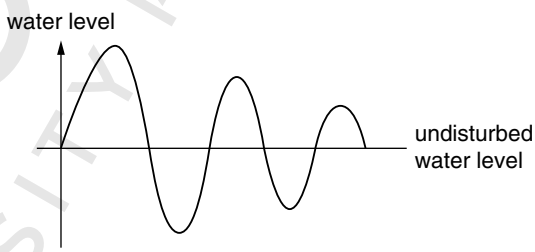


Fig ad

(c) Q is a point on the line joining S_1 and S_2 as shown in Figure ac. State the kind of interference that occurs at Q and give a reason for this occurrence. (2 marks)

31 HKDSE 2017 Paper 1B Q6

5.3

(a) A dipper vibrating with a frequency of 5 Hz is put in a water tank. Figure ae shows the displacement–distance graph of the water wave at time $t = 0$. Y is a particle in the water tank.

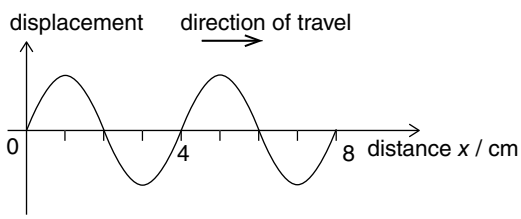


Fig ae

(i) Determine the wave speed of the water wave. (2 marks)

(ii) State the direction of motion of particle Y at $t = 0$. (1 mark)

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(iii) Sketch the displacement–time graph of particle Y between $t = 0$ and $t = 0.4$ s in Figure af. (2 marks)

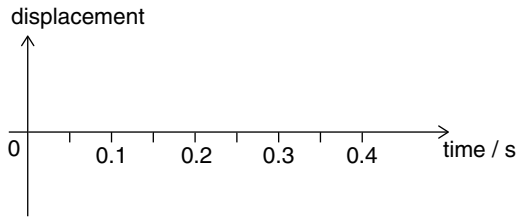


Fig af

(b) In figure ag, A and B are two dippers vibrating in phase in a water tank. The distance between A and B is 6 cm. OP is the perpendicular bisector of AB. Q is a second minimum from P, where $AQ = 12$ cm and $BQ = 15$ cm.

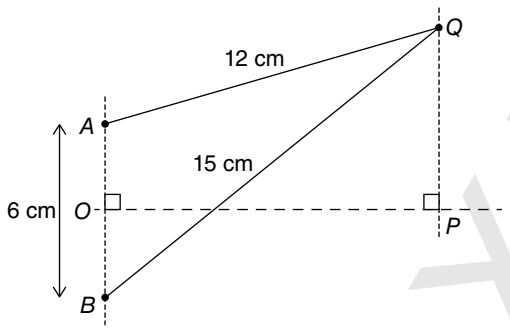


Fig ag

(i) Explain why a minimum occurs at Q. (2 marks)

(ii) Determine the wavelength of the water wave. (2 marks)

(iii) Sketch in Figure ah how the **amplitude** of the water wave varies along the line OP. (1 mark)

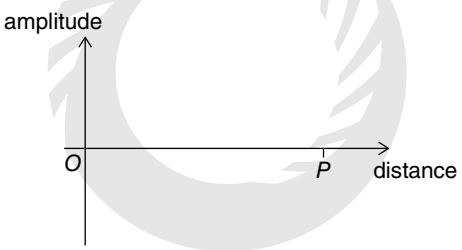


Fig ah

Q31 Exam report: This question tested candidates' basic knowledge of wave motion. Candidates' performance was satisfactory. Part (a) was well answered. In (b)(i), some candidates only stated 'crest meets trough at Q' to explain why a minimum occurred there. Quite a number of them did not know that in (b)(iii) the amplitude of the water waves would decrease gradually with the distance from the source.

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Misc

A ripple tank has a shallow region P and a deep region Q. Straight water wave of frequency 10 Hz is travelling in the shallow region as shown in Figure ai when viewed from above.

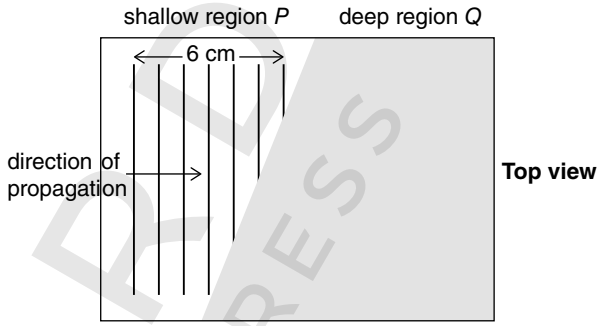


Fig ai

(a) The separation between seven crests in the shallow region is found to be 6 cm as shown.

(i) Find the wavelength of the wave in the shallow region. (1 mark)

(ii) What is the wave speed in the shallow region? (1 mark)

(b) The water wave then propagates into the deep region where the wavelength of the wave is double that in the shallow region.

(i) State the frequency of the water wave in the deep region. (1 mark)

(ii) On Figure ai, sketch the wave pattern in the deep region. (2 marks)

(iii) Name the phenomenon occurred across the boundary and explain its cause. (2 marks)

Q32 Exam report: This question was in general well answered. Most obtained the correct answers in (a) although a few wrongly counted the number of wavelengths in the shallow region. In (b)(i), the less able candidates struggled to find the frequency of the water wave in the deep region, which should remain unchanged (at 10 Hz). Quite a number of the candidates failed to sketch the refracted wavefronts properly in (b)(ii). Common mistakes included wavefronts bending towards instead of away from the normal or not continuing across the boundary. Part (b) (iii) was well answered except for a few who confused refraction with diffraction.

Know more

Ripple tank (p.39)

A ripple tank has some special features to facilitate the production and observation of water waves.

- The motor has an off-centred weight attached to the axle (Fig a). It wobbles when turned on to make the bar (or dipper) vibrate.
- The sides of the tank are lined with a layer of sponge, or they are inclined instead of being vertical (Fig b). This reduces the bouncing back of water waves from the sides.



Fig a

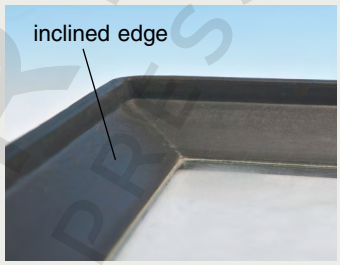


Fig b

Patterns of stationary waves can also be formed using a vibrator, a metal plate and sand.

Video

→ This video shows the wave patterns formed on a circular plate at particular frequencies.

Stationary waves on a surface (p.79)

Stationary waves also form on a surface. If a dish of water is fixed to a vibrating surface, such as a speaker, special patterns corresponding to different frequencies are formed (Fig c).

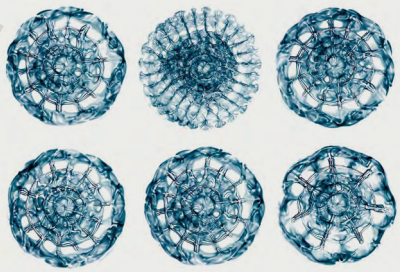


Fig c

Video

→ This video shows how a paper rider is thrown out when tuning a guitar.

Tuning a guitar (p.80)

When a vibrating tuning fork is put on a guitar's body, it makes the strings vibrate with the same frequency. If this frequency is the same as the frequency that allows a stationary wave of 1 loop to be formed on a string, the string will vibrate with a large amplitude and cause the paper rider to be thrown out (Fig d).

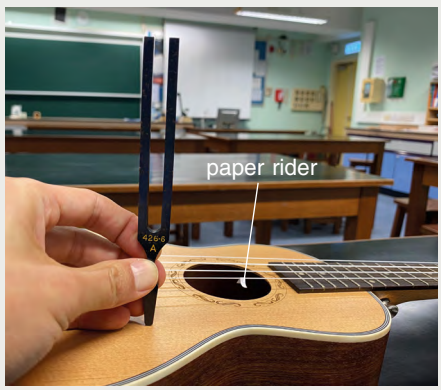


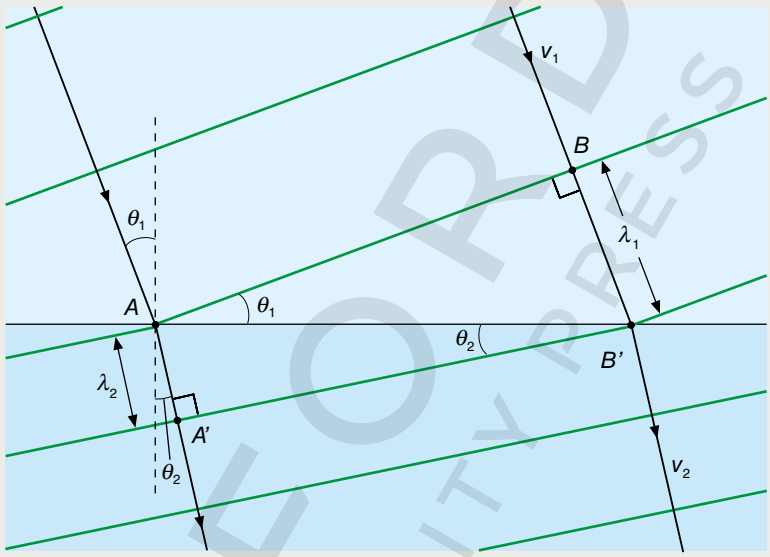
Fig d

Q30 Exam Report: Part (a) was poorly answered. Not many candidates identified the correct positions of the antinodal line L_1 and nodal line L_2 by drawing a smooth curve through the points for each line. Many candidates wrongly drew straight lines for L_1 and L_2. Only the more able ones correctly pointed out that the separation between L_1 and L_2 would increase when S_1S_2 was reduced slightly in separation. Candidates' performance in (b) and (c) was satisfactory. Many failed to give the correct path difference at Q.

Know more

Derivation of Snell's law (p.53)

The figure below shows a straight wave undergoing refraction when it crosses from region 1 with wave speed v_1 to region 2 with wave speed v_2. The wavelengths of the wave in regions 1 and 2 are λ_1 and λ_2 respectively. The angle of incidence is θ_1 and the angle of refraction is θ_2.



During refraction, f remains unchanged.

$f_1 = f_2$

By $v = f\lambda$, we have

$$\frac{v_1}{\lambda_1} = \frac{v_2}{\lambda_2}$$

$$\frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2} \quad \text{..... (1)}$$

Consider the area bounded by $AA'B'B$. The angle between the incident wavefront and the boundary is equal to θ_1 (see DSE practice 1 on p.55). Similarly, the angle between the refracted wavefront and the boundary is equal to θ_2.

Consider right angle triangle $\triangle ABB'$,

$$\sin\theta_1 = \frac{\lambda_1}{AB'} \quad \text{..... (2)}$$

Consider right angle triangle $\triangle AA'B'$,

$$\sin\theta_2 = \frac{\lambda_2}{AB'} \quad \text{..... (3)}$$

Dividing (2) by (3), we have

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{\lambda_1}{\lambda_2} \quad \text{..... (4)}$$

By comparing (1) and (4), we obtain

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$